

Concept Revision Notes

Everything you need to know to solve the questions in **School Paper Based** — the Heritage School's two Mathematics papers and the similar sums that go with them — arranged chapter by chapter.

Nothing here is general revision. Every item is on the page because one or more of those questions needs it, and each says which. Where a concept has a standard way of going wrong, that is set out too.

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1. Relations and Functions

Four questions, and between them they test every definition in the chapter. Nothing here needs calculation — it is all about stating a definition precisely and then checking it.

1.1 The three properties, with their quantifiers

Reflexive: $(a, a) \in R$ for **every** a in the set. **Symmetric:** whenever $(a, b) \in R$, also $(b, a) \in R$. **Transitive:** whenever (a, b) and (b, c) are both in R , so is (a, c) .

WATCH OUT Reflexive needs *every* element, so one missing pair kills it. Symmetric and transitive are conditional — they only ask for something when the premise holds.

NEEDED FOR FT Q1(xv) FT Q8

1.2 Vacuous truth

If a relation contains no pair (a, b) and (b, c) to trigger transitivity, the condition is satisfied **vacuously** and the relation *is* transitive.

WATCH OUT This is the commonest one-mark trap in the chapter. $R = \{(1, 2)\}$ is transitive; $R = \{(2, 2)\}$ is symmetric *and* transitive.

NEEDED FOR FT Q1(xv) CISCE competency booklet, Q28

1.3 Equivalence relation and its classes

All three properties together. The classes partition the set, and two elements are in the same class exactly when they are related.

WATCH OUT A parity condition is transitive; a **bounded-difference** condition such as $|a - b| \leq 5$ is not — differences add but the bound does not.

NEEDED FOR FT Q8 NCERT Part I, Chapter 1

1.4 One-one, onto, bijection — how each is proved

One-one: assume $f(x_1) = f(x_2)$ and derive $x_1 = x_2$. **Onto:** take any y in the co-domain and solve $y = f(x)$ for an x in the domain.

NEEDED FOR FT Q1(vi) NCERT Exemplar Ch 1, Q21(iii)

1.5 Strictly increasing $\Rightarrow$ one-one

If f is strictly increasing (or strictly decreasing) on an interval it is automatically one-one there. With continuity, the intermediate value theorem then makes it onto its range.

WATCH OUT $x|x|$ is strictly increasing across 0 because the modulus keeps the sign; x^2 is not.

NEEDED FOR FT Q1(vi)

1.6 Range against co-domain

The co-domain is where the function is *declared* to land; the range is what it actually reaches. A question asking for the range wants the second.

NEEDED FOR FT Q1(ii) NCERT Exemplar Ch 1, Q14

1.7 Finding a range by completing the square

$f(x) = x^2 - 4x + 5 = (x - 2)^2 + 1$. On a restricted domain, read off what the bracket can do and push it through.

WATCH OUT The restriction to $[2, \infty)$ is what makes the function one-one; without it the same formula is many-one.

NEEDED FOR FT Q1(ii)

2. Inverse Trigonometric Functions

Five questions. Every one of them turns on the **principal value branch** — the algebra is easy, and the marks are in knowing when an identity is allowed.

2.1 The principal value branches

FUNCTION	BRANCH
$\sin^{-1} x$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[0, \pi]$
$\tan^{-1} x$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1} x$	$(0, \pi)$

WATCH OUT Nothing else in this chapter matters as much. Learn the four.

NEEDED FOR FT Q1(iii) FT Q7 PB Q3(b) NCERT Exemplar Ch 2, Q5

2.2 The two complementary identities

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \quad \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

WATCH OUT Whenever two different inverse functions of the *same* x appear, reach for these first — they usually collapse the whole expression.

NEEDED FOR FT Q1(iii) FT Q3 NCERT Exemplar Ch 2, Q22 NCERT Exemplar Ch 2, Q26

2.3 Folding an angle back into its branch

$f^{-1}(f(\theta)) = \theta$ **only** when θ is in the branch. Outside it, subtract the right period: π for $\tan^{-1}$, 2π for $\sin^{-1}$ and $\cos^{-1}$.

WATCH OUT Using 2π on a tangent is the usual error. The answer to FT Q7 is $\frac{9\pi}{4} - x$, *not* $\frac{\pi}{4} - x$, purely because of the interval given.

NEEDED FOR FT Q7 NCERT Exemplar Ch 2, Q5

2.4 Converting one inverse function into another

Draw the right triangle. $\sin^{-1}\frac{4}{5}$ has opposite 4 and hypotenuse 5, so adjacent 3 and the angle is also $\tan^{-1}\frac{4}{3}$.

WATCH OUT Only valid for the acute case; check the sign of the cosine before assuming it.

NEEDED FOR PB Q1(b) NCERT Exemplar Ch 2, Q14

2.5 The $\tan^{-1}$ addition formula, and its condition

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1} \frac{x+y}{1-xy}, \quad xy < 1$$

WATCH OUT When $xy > 1$ a $\pm\pi$ must be added — that is what the condition is protecting you from.

NEEDED FOR PB Q1(b) NCERT Part I, Chapter 2

2.6 Naming the angles, then expanding

For $\sin^{-1}A \pm \sin^{-1}B$, set $A = \sin\alpha$, $B = \sin\beta$, read $\cos\alpha$ and $\cos\beta$ off the triangles, and expand $\sin(\alpha \pm \beta)$.

WATCH OUT Confirm the sign of $\cos(\alpha \pm \beta)$ before quoting the answer as an inverse sine — the obtuse angle has the same sine.

NEEDED FOR PB Q3(b) NCERT Exemplar Ch 2, Q14

2.7 Compressing a $\sin x + b \cos x$

$$\frac{\sin x + \cos x}{\sqrt{2}} = \sin\left(x + \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4} - x\right). \text{ In general } a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \phi).$$

NEEDED FOR FT Q7 NCERT Exemplar Ch 5, Q37

2.8 Test every candidate root

Squaring or cross-multiplying inside an inverse-trig equation can manufacture roots that the branches then reject.

WATCH OUT In PB Q3(b) the algebra gives $x = 2$ and $x = -1$; only $x = 2$ survives.

NEEDED FOR PB Q3(b) NCERT Exemplar Ch 2, Q7

3. Matrices

Four questions: one on transposes, one decomposition, and two systems of equations solved by the inverse.

3.1 Transpose rules

$(A')' = A$, $(A + B)' = A' + B'$, and — the one that matters — $(AB)' = B'A'$, with the order reversed.

WATCH OUT Keeping the order gives the wrong sign and therefore the wrong answer.

NEEDED FOR FT Q1(i) NCERT Exemplar Ch 3, Q63

3.2 Symmetric and skew symmetric

$A' = A$ is symmetric; $A' = -A$ is skew. A skew matrix always has a **zero diagonal**, because $a_{ii} = -a_{ii}$.

WATCH OUT The zero diagonal is the quickest check that an answer is right.

NEEDED FOR FT Q1(i) PB Q1(a)

3.3 The unique split

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

WATCH OUT First bracket symmetric, second skew; the decomposition exists for every square matrix and is unique.

NEEDED FOR PB Q1(a) NCERT Part I, Chapter 3, Exercise 3.3 NCERT Exemplar Ch 3, Q52

3.4 Multiplication does not commute

$AB \neq BA$ in general, which is why $(A + B)^2 = A^2 + 2AB + B^2$ holds only when $AB = BA$.

NEEDED FOR FT Q1(i)

3.5 Inverse by the adjoint

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad |A| \neq 0$$

WATCH OUT **adj A is the transpose of the cofactor matrix.** Forgetting the transpose is the single commonest way to lose these marks.

NEEDED FOR PB Q2(b) FT Q11

3.6 Solving a system by the matrix method

Write $AX = B$, check $|A| \neq 0$, then $X = A^{-1}B$. A unique solution exists precisely when $|A| \neq 0$.

WATCH OUT Always substitute the answer back into all three equations — it costs ten seconds and catches every arithmetic slip.

NEEDED FOR FT Q11 PB Q2(b) NCERT Exemplar Ch 4, Q19 NCERT Part I, Chapter 4

3.7 Turning a word problem into a system

One variable per unknown price or quantity, one equation per purchase or condition. The matrix work is then mechanical.

NEEDED FOR FT Q11 NCERT Part I, Chapter 4

4. Determinants

Four questions, none of which should be expanded by brute force. Every one is a properties question.

4.1 The properties worth knowing by name

- A common factor in a row or column comes out.
- Two identical — or proportional — rows or columns make it zero.
- $R_i \rightarrow R_i + kR_j$ leaves the value unchanged.
- Interchanging two rows changes the sign.

NEEDED FOR FT Q1(iv) FT Q6 PB Q2(a) NCERT Exemplar Ch 4, Q6 NCERT Exemplar Ch 4, Q7 NCERT Exemplar Ch 4, Q13

4.2 The standard opening move

$R_1 \rightarrow R_1 + R_2 + R_3$ (or the same with columns) often makes a whole row constant. Take that factor out, then clear the row with two more operations.

WATCH OUT Two operations are usually needed after the factor comes out; stopping after one leaves a determinant you still have to expand.

NEEDED FOR PB Q2(a) FT Q1(iv) NCERT Exemplar Ch 4, Q13

4.3 Determinants of products and powers

$$|AB| = |A||B|, \quad |A^n| = |A|^n, \quad |kA| = k^n|A| \text{ for an } n \times n \text{ matrix}$$

WATCH OUT $|kA| = k^n|A|$: the exponent is the **order**, not 1.

NEEDED FOR FT Q4

4.4 Adjoint identities

$$A (\text{adj } A) = |A| I_n, \quad |\text{adj } A| = |A|^{n-1}$$

NEEDED FOR FT Q4

4.5 Checking a determinant identity cheaply

Substitute convenient numbers — $a = b = c = 1$, or $x = 0$ — into both sides. It will not prove the identity but it catches a wrong constant at once.

NEEDED FOR FT Q1(iv) PB Q2(a)

4.6 Spotting a linear relation between columns

If one column is a combination of the others the determinant is zero. Look for it before expanding: $(p + q)^2 - (p - q)^2 = 4pq$ is what makes FT Q6 vanish.

NEEDED FOR FT Q6 NCERT Exemplar Ch 4, Q7

5. Continuity and Differentiability

The heaviest chapter in both papers — eight questions. Half of them are ‘prove this second-order identity’, which is one template applied four times.

5.1 Continuity at a point

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

WATCH OUT All three parts, not two: the limits can agree with each other and still differ from the value.

NEEDED FOR FT Q1(ix) FT Q12 NCERT Exemplar Ch 5, Q11 NCERT Exemplar Ch 5, Q16

5.2 Differentiability implies continuity

A function differentiable at a point is continuous there. The converse is false — $|x|$ at 0 is the standard counter-example.

WATCH OUT So a ‘derivable’ question carries **two** conditions: match the values and match the one-sided slopes.

NEEDED FOR FT Q12 CISCE competency booklet, Q22

5.3 Piecewise functions: two conditions

At the join, continuity gives one equation and equality of the one-sided derivatives gives another. Two unknowns need both.

WATCH OUT Matching only the slopes leaves the constant free and the function able to jump.

NEEDED FOR FT Q12 NCERT Exemplar Ch 5, Q16

5.4 Parametric first derivative

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

NEEDED FOR FT Q1(vii) FT Q1(viii) NCERT Exemplar Ch 6, Q44

5.5 Parametric second derivative — the trap

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

WATCH OUT It is **not** $\frac{d^2y}{dt^2} \div \frac{d^2x}{dt^2}$. FT Q1(viii) is built so that the wrong rule gives the assertion's wrong answer — both halves of it are false.

NEEDED FOR FT Q1(viii)

5.6 Simplify before you differentiate

$e^{\log t} = t$ for $t > 0$, so $y = e^{\log \sin^{-1} x} + e^{\log \cos^{-1} x}$ is just $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

WATCH OUT A constant differentiates to 0. Grinding through the chain rule gets there too, but slowly and with two chances to slip.

NEEDED FOR FT Q2 NCERT Exemplar Ch 5, Q37

5.7 Logarithmic differentiation

Take logs of both sides first when the unknown appears in an exponent or as a product of many factors.

WATCH OUT $(ax + b)e^{\frac{y}{x}} = x$ becomes $y = x \log x - x \log(ax + b)$ in one line.

NEEDED FOR FT Q9

5.8 The second-order identity template

1. Differentiate once and rearrange so the awkward root sits alone.
2. Square both sides to clear it.
3. Differentiate again.
4. Divide through by $2 \frac{dy}{dx}$, noting it is not identically zero.

WATCH OUT This one template answers every 'show that $(1 \pm x^2)y'' \pm xy' = \dots$ ' question in either paper.

NEEDED FOR FT Q9(OR) PB Q5(a) NCERT Exemplar Ch 5, Q81

5.9 Standard limits and small-angle expansions

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \sin x = x - \frac{x^3}{6} + \dots, \quad \cos x = 1 - \frac{x^2}{2} + \dots$$

WATCH OUT Two terms of each series settle a $\frac{0}{0}$ limit faster than three applications of L'Hôpital.

NEEDED FOR PB Q1(d)

5.10 Derivatives of the inverse trigonometric functions

$$\frac{d}{dx} \sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \tan^{-1}x = \frac{1}{1+x^2}$$

NEEDED FOR FT Q2 NCERT Exemplar Ch 5, Q37

6. Applications of Derivatives

Eight questions: two rates, three tangent-and-normal, three optimisation. The optimisation ones all have a constraint that must be used to remove a variable.

6.1 Related rates

Differentiate the relation with respect to **t**, then substitute the instantaneous values. The rate asked for is the one on the left; everything else is data.

NEEDED FOR NCERT Exemplar Ch 6, Q35 NCERT Exemplar Ch 6, Q3

6.2 A rate with respect to something other than time

FT Q14 asks for $\frac{dV}{dh}$, not $\frac{dV}{dt}$. Express V in terms of that one variable and differentiate once.

WATCH OUT No $\frac{dh}{dt}$ is given and none is needed — read which variable the question actually names.

NEEDED FOR FT Q14

6.3 Increasing and decreasing from the sign of f'

$f' > 0$ on an interval $\Rightarrow$ strictly increasing there. Factorise f' and read the sign off the roots.

WATCH OUT A positive-leading quadratic is negative **between** its roots; a negative-leading one is positive between them.

NEEDED FOR FT Q1(x) NCERT Exemplar Ch 6, Q46

6.4 Tangent and normal

Slope of the tangent is $\frac{dy}{dx}$ at the point; the normal's slope is $-1 \div$ that. Then use $y - y_1 = m(x - x_1)$.

WATCH OUT $-\frac{1}{m}$, not $-m$.

NEEDED FOR FT Q5 FT Q5(OR) NCERT Exemplar Ch 6, Q41 NCERT Exemplar Ch 6, Q14

6.5 'The tangent is this particular line'

Matching the slope is **necessary but not sufficient**. Find every point with that slope, then check which of them actually lies on the given line.

WATCH OUT In FT Q5 two points have slope 1 and only one lies on $y = x - 11$.

NEEDED FOR FT Q5

6.6 Implicit differentiation

Differentiate both sides as they stand, treating y as a function of x , then solve for $\frac{dy}{dx}$.

WATCH OUT $\sqrt{x} + \sqrt{y} = 4$ gives $\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$ — always negative, which decides the sign when the tangent is ‘equally inclined’.

NEEDED FOR FT Q5 (OR) NCERT Exemplar Ch 6, Q14

6.7 Maxima and minima: the two tests

First derivative: f' changes $+$ to $-$ at a maximum. **Second derivative:** $f' = 0$ and $f'' < 0$ gives a maximum.

WATCH OUT If f'' is also zero the second test says nothing — fall back on the sign of f' .

NEEDED FOR PB Q5(b) FT Q14(OR) NCERT Exemplar Ch 6, Q31

6.8 On a closed interval, test the endpoints

The absolute maximum of a continuous function on $[a, b]$ is the largest of the values at the stationary points **and** at a and b .

WATCH OUT In FT Q1(v) both stationary points lie outside $[-1, 1]$, so the answer is at an endpoint and solving $f' = 0$ alone finds nothing.

NEEDED FOR FT Q1(v)

6.9 Constrained optimisation

Use the constraint to eliminate one variable *before* differentiating, or differentiate the constraint to get $\frac{dr}{da}$ and substitute.

WATCH OUT Treating the two variables as independent and setting both derivatives to zero collapses the problem to nothing.

NEEDED FOR PB Q5(b) FT Q14(OR) NCERT Exemplar Ch 6, Q31 NCERT Part I, Chapter 6

7. Integrals

Only three questions in the two papers — the 2024-25 paper has no integration at all — but all three are property-and-identity questions rather than routine ones.

7.1 The half-angle identities

$$1 - \cos x = 2 \sin^2 \frac{x}{2}, \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}, \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

WATCH OUT These turn an apparently hopeless quotient into cosec^2 and $\cot$, which integrate at sight.

NEEDED FOR PB Q1(e) NCERT Part II, Chapter 7

7.2 $\frac{f'(x)}{f(x)}$ integrates to $\log|f(x)|$

Before any substitution, check whether the numerator is the derivative of the denominator.

NEEDED FOR PB Q1(e) NCERT Part II, Chapter 7

7.3 Integration by parts, and the cancellation pattern

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

WATCH OUT In PB Q1(e) the term parts produces is exactly the one already waiting to be integrated, and the two cancel. Look for that before grinding on.

NEEDED FOR PB Q1(e)

7.4 The $a - x$ reflection property

$$\int_0^a f(x) dx = \int_0^a f(a - x) dx$$

WATCH OUT On $[0, \frac{\pi}{2}]$ it swaps $\sin$ and $\cos$, and $\tan$ and $\cot$. Adding the two forms usually gives $2I = \text{something trivial}$.

NEEDED FOR PB Q1(f) NCERT Exemplar Ch 7, Q30

7.5 Rewrite before you substitute

$\frac{1}{1 + \tan x} = \frac{\cos x}{\cos x + \sin x}$ is what exposes the symmetry; in that form the property works, in the original it does not.

NEEDED FOR PB Q1(f) NCERT Part II, Chapter 7, Example 6(iii)

7.6 Combining $\sqrt{\tan x}$ and $\sqrt{\cot x}$

Put them over a common denominator to get $\frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}$, then use

WATCH OUT $2 \sin x \cos x = 1 - (\sin x - \cos x)^2$ — not $\sin 2x$, which leaves a root you cannot substitute into.

NEEDED FOR PB Q6(a) NCERT Exemplar Ch 7, Q43

7.7 Changing the limits with the variable

In a definite integral, either convert the limits when you substitute or convert back before evaluating. Never mix the two.

NEEDED FOR PB Q1(f)

8. Application of Integrals

One question in the two papers, but it needs the whole method — and the first step is not integration at all.

8.1 Identify the curve first

$y^2 = 2x - x^2$ rearranges to $(x - 1)^2 + y^2 = 1$: a **circle** of radius 1 centred at (1, 0).

WATCH OUT Complete the square before anything else. Integrating blindly turns a quarter circle into an intractable integral.

NEEDED FOR PB Q6(b) NCERT Exemplar Ch 8, Q16

8.2 Area between two curves

$$A = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) dx$$

WATCH OUT Decide which curve is on top by testing one convenient x between the intersections.

NEEDED FOR PB Q6(b) NCERT Exemplar Ch 8, Q16

8.3 Split where the boundary changes

If the upper curve changes, or a modulus changes definition, the integral splits at that point.

NEEDED FOR NCERT Exemplar Ch 8, Q18 NCERT Exemplar Ch 8, Q23

8.4 Area is not the signed integral

Where the curve dips below the axis the integral is negative. An 'area' answer of zero always means a missing modulus.

NEEDED FOR NCERT Exemplar Ch 8, Q17

8.5 Standard areas worth recognising

- A quarter circle of radius a has area $\frac{\pi a^2}{4}$.
- A sector of angle θ has area $\frac{1}{2}r^2\theta$.
- Integrating along y is often shorter when both curves give x in terms of y .

NEEDED FOR PB Q6(b) NCERT Exemplar Ch 8, Q27 NCERT Exemplar Ch 8, Q29

9. Linear Programming

Five questions — and the two long ones are the **same diet problem**, set eight years apart with the grains and prices exchanged. Formulation is most of the marks.

9.1 Formulating

1. Name the variables, with their units.
2. One inequality per resource or requirement.
3. Write the objective, and say whether it is to be maximised or minimised.
4. Add $x, y \geq 0$.

WATCH OUT **Convert the units first.** Costs quoted per kg against contents quoted per gram is exactly where these questions catch people.

NEEDED FOR FT Q22 PB Q13(b) NCERT Exemplar Ch 12, Q23

9.2 Half planes, strict and non-strict

A linear inequality is a half plane. A strict one ($<$) excludes its boundary; test the origin to see which side is meant.

NEEDED FOR FT Q19(ii) NCERT Exemplar Ch 12

9.3 The corner point theorem

On a **bounded** feasible region the optimum of a linear objective is attained at a corner point. Evaluate Z at each and compare.

NEEDED FOR FT Q19(v) FT Q20 NCERT Exemplar Ch 12, Q1 NCERT Exemplar Ch 12, Q10

9.4 Unbounded regions need one more check

If the region is unbounded, a corner value M is the minimum only if the open half plane $ax + by < M$ has **no point in common** with the region.

WATCH OUT Quoting the smallest corner value without this check is an incomplete answer — a minimum need not exist at all.

NEEDED FOR FT Q22 PB Q13(b)

9.5 Ties between two corners

If Z takes the same value at two corner points, the objective line is parallel to that edge and **every** point of the edge is optimal.

NEEDED FOR FT Q19(v) NCERT Exemplar Ch 12, Q22

9.6 Feasible solutions against the optimal solution

Every point of the region is a feasible solution — so there are usually infinitely many. Only the *optimum* is confined to the corners.

WATCH OUT FT Q20 asks for the number of feasible solutions, not the maximum. The answer is 'infinite'.

NEEDED FOR FT Q20

9.7 Not every pair of lines gives a corner

Two constraint lines may meet outside the feasible region, and a constraint may be redundant. Test each intersection against *all* the constraints.

NEEDED FOR NCERT Exemplar Ch 12, Q21

Appendix — every problem referred to, with its solution

FT Q1(xv) — Heritage 2024–25 First Term

A relation R on the set $\{1, 2, 3\}$ is defined by $R = \{(1, 2)\}$. Classify R .

- (a) reflexive
- (b) transitive
- (c) symmetric
- (d) none of these

SOLUTION

1. Reflexive would need $(1,1)$, $(2,2)$ and $(3,3)$; none is there.
2. Symmetric would need $(2,1)$ to accompany $(1,2)$; it is not there.
3. Transitive asks: whenever (a,b) and (b,c) lie in R , does (a,c) ? The only pair is $(1,2)$, and there is no pair starting at 2, so the condition is never triggered.

NOTE A condition that is never triggered is satisfied **vacuously**. This is the standard trap: a one-element relation is almost always transitive.

ANSWER transitive (vacuously), but neither reflexive nor symmetric

FT Q8 — Heritage 2024–25 First Term

Test whether the relation R on $\mathbb{Z}$ defined by $R = \{(a, b) : |a - b| \leq 5\}$ is reflexive, symmetric and transitive.

SOLUTION

1. **Reflexive.** $|a - a| = 0 \leq 5$ for every integer a , so $(a, a) \in R$. R is reflexive.
2. **Symmetric.** $|a - b| = |b - a|$, so $(a, b) \in R$ forces $(b, a) \in R$. R is symmetric.
3. **Transitive.** Take $a = 1$, $b = 6$, $c = 11$. Then $|1 - 6| = 5 \leq 5$ and $|6 - 11| = 5 \leq 5$, so $(1,6)$ and $(6,11)$ lie in R .
4. But $|1 - 11| = 10 > 5$, so $(1, 11) \notin R$. R is not transitive.

NOTE One counter-example is enough to kill transitivity, and it must be produced explicitly — saying ‘the gaps can add up’ earns nothing.

TRAP A bounded-difference relation is never transitive: differences add, but the bound does not.

ANSWER Reflexive and symmetric, but **not** transitive

CISCE competency booklet, Q28 — CISCE competency booklet

Statement II: the relation R on the set $\{2, 3, 4\}$ defined by $R = \{(2, 2)\}$ is not symmetric.

SOLUTION

1. Symmetry asks: if $(a, b) \in R$, is (b, a) ? The only pair is $(2, 2)$, whose reverse is itself.
2. So R **is** symmetric — and transitive too, vacuously — though not reflexive, since $(3, 3)$ and $(4, 4)$ are missing.

→ Statement II is **false**

WHY IT IS SET HERE That statement is **false**: $R = \{(2, 2)\}$ is symmetric, and transitive, though not reflexive. CISCE sets the same one-pair relation.

NCERT Part I, Chapter 1 — NCERT

Show that $R = \{(a, b) : |a - b| \text{ is even}\}$ on $\{1, 2, 3, 4, 5\}$ is an equivalence relation.

SOLUTION

1. **Reflexive**: $|a - a| = 0$ is even. **Symmetric**: $|a - b| = |b - a|$.
 2. **Transitive**: if $a - b$ and $b - c$ are both even, so is their sum $a - c$.
- An equivalence relation; the classes are $\{1, 3, 5\}$ and $\{2, 4\}$ — the two parities

WHY IT IS SET HERE It is — and the contrast is the point: a **parity** condition is transitive, a **bounded-difference** condition like the school's $|a - b| \leq 5$ is not.

FT Q1(vi) — Heritage 2024–25 First Term

Let $A = \{x : -1 \leq x \leq 1\}$ and $f : A \rightarrow A$ be defined by $f(x) = x|x|$. Then f is:

- (a) a bijection
- (b) injective but not surjective
- (c) surjective but not injective
- (d) neither injective nor surjective

SOLUTION

1. Write f piecewise: $f(x) = x^2$ for $x \geq 0$ and $f(x) = -x^2$ for $x < 0$.
2. Both branches are strictly increasing, and they meet at $f(0) = 0$, so f is strictly increasing on the whole of $[-1, 1]$ — hence one-one.
3. $f(-1) = -1$ and $f(1) = 1$, and f is continuous, so by the intermediate value theorem every value in $[-1, 1]$ is taken. f is onto A .

TRAP $x|x|$ is **not** x^2 . The modulus keeps the sign, which is exactly what makes this one-one where x^2 would not be.

ANSWER a bijection — f is strictly increasing on A and its range is all of A

NCERT Exemplar Ch 1, Q21(iii) — NCERT Exemplar

On $A = [-1, 1]$, discuss whether $h(x) = x|x|$ is one-one, onto or bijective.

SOLUTION

1. $h(x) = x^2$ for $x \geq 0$ and $-x^2$ for $x < 0$, so h is strictly increasing across the whole of $[-1, 1]$ — hence one-one.
2. $h(-1) = -1$, $h(1) = 1$ and h is continuous, so every value between is taken.
→ Bijective

WHY IT IS SET HERE Bijective — **the identical question**, which is where the school took it from.

FT Q1(ii) — Heritage 2024–25 First Term

Let $f : [2, \infty) \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 - 4x + 5$. Then the range of f is:

- (a) $\mathbb{R}$
- (b) $[1, \infty)$
- (c) $[4, \infty)$
- (d) $[5, \infty)$

SOLUTION

1. Complete the square: $f(x) = (x - 2)^2 + 1$.
2. On $[2, \infty)$ the bracket $(x - 2)$ runs over $[0, \infty)$, so $(x - 2)^2$ runs over $[0, \infty)$.
3. Hence $f(x) = (x - 2)^2 + 1$ takes every value from 1 upwards, and 1 is attained at $x = 2$.

NOTE The co-domain is $\mathbb{R}$, but the *range* is what the function actually reaches. Restricting the domain to $[2, \infty)$ is what makes f one-one here.

ANSWER $[1, \infty)$

NCERT Exemplar Ch 1, Q14 — NCERT Exemplar

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = \frac{1}{2 - \cos x}$. Find the range of f .

SOLUTION

1. $\cos x$ sweeps $[-1, 1]$, so $2 - \cos x$ sweeps $[1, 3]$ and is never zero.
2. Reciprocating reverses a positive interval:
→ Range = $[\frac{1}{3}, 1]$, both ends attained

WHY IT IS SET HERE $[\frac{1}{3}, 1]$ — *same method: read off the range of the inside, then push it through the outside.*

FT Q1(iii) — Heritage 2024–25 First Term

If $a \leq \cos^{-1} x + 2 \sin^{-1} x \leq b$, then:

- (a) $a = -\frac{\pi}{2}, b = \frac{\pi}{2}$
 (b) $a = 0, b = \pi$
 (c) $a = -\frac{\pi}{2}, b = \frac{3\pi}{2}$
 (d) $a = 0, b = 2\pi$

SOLUTION

1. Use $\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$ to collapse the expression:

$$\cos^{-1}x + 2 \sin^{-1}x = \frac{\pi}{2} + \sin^{-1}x$$

2. $\sin^{-1}x$ runs over $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

3. Adding $\frac{\pi}{2}$ shifts that to $[0, \pi]$.

NOTE Whenever both $\cos^{-1}$ and $\sin^{-1}$ of the *same* x appear, add and subtract to reduce to one of them.

ANSWER $a = 0, b = \pi$

FT Q7 — Heritage 2024–25 First Term

Find the value of $\cos^{-1}[\frac{\sin x + \cos x}{\sqrt{2}}]$, where $\frac{5\pi}{4} < x < \frac{9\pi}{4}$.

SOLUTION

1. Compress the numerator: $\frac{\sin x + \cos x}{\sqrt{2}} = \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} = \sin(x + \frac{\pi}{4})$.

2. Now $\sin\theta = \cos(\frac{\pi}{2} - \theta)$, so the expression is $\cos^{-1}[\cos(\frac{\pi}{4} - x)]$.

3. For $\cos^{-1}(\cos\phi) = \phi$ we need $\phi \in [0, \pi]$. Here $x \in (\frac{5\pi}{4}, \frac{9\pi}{4})$ gives $\frac{\pi}{4} - x \in (-2\pi, -\pi)$, which is not that interval.

4. Add 2π (cosine is 2π -periodic): $\phi = \frac{9\pi}{4} - x$, which lies in $(0, \pi)$. ✓

TRAP Writing the answer as $\frac{\pi}{4} - x$ scores nothing: it is negative on the given interval, and $\cos^{-1}$ never is. The range restriction is the whole question.

ANSWER $\cos^{-1}[\frac{\sin x + \cos x}{\sqrt{2}}] = \frac{9\pi}{4} - x$

PB Q3(b) — Heritage 2016–17 Pre-Board

Solve for x : $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \sin^{-1}\left(\frac{1+x}{1+x^2}\right)$.

SOLUTION

NOTE Name the two angles and read their sines and cosines straight off a right triangle with legs x and 1 and hypotenuse $\sqrt{1+x^2}$.

1. Let $A = \sin^{-1}\frac{x}{\sqrt{1+x^2}}$ and $B = \sin^{-1}\frac{1}{\sqrt{1+x^2}}$.

2. Then $\sin A = \frac{x}{\sqrt{1+x^2}}$, $\cos A = \frac{1}{\sqrt{1+x^2}}$, $\sin B = \frac{1}{\sqrt{1+x^2}}$, $\cos B = \frac{x}{\sqrt{1+x^2}}$.

3. **Subtract.**

$$\sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{x^2 - 1}{1 + x^2}$$

4. So the equation reads $\frac{x^2 - 1}{1 + x^2} = \frac{1 + x}{1 + x^2}$, i.e. $x^2 - 1 = 1 + x$.

5. $x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = 2$ or $x = -1$.

TRAP Test both roots. At $x = -1$ the left side is $-\frac{\pi}{4} - \frac{\pi}{4} = -\frac{\pi}{2}$ while the right side is $\sin^{-1}0 = 0$. It fails. Only $x = 2$ survives — the algebra generated it, the principal-value ranges reject it.

ANSWER $x = 2$

NCERT Exemplar Ch 2, Q5 — NCERT Exemplar

Find the value of $\tan^{-1}(\tan \frac{2\pi}{3})$.

SOLUTION

1. $\frac{2\pi}{3}$ lies outside $(-\frac{\pi}{2}, \frac{\pi}{2})$, so subtract the period π of the tangent:

$$\tan \frac{2\pi}{3} = \tan\left(\frac{2\pi}{3} - \pi\right) = \tan\left(-\frac{\pi}{3}\right)$$

2. and $-\frac{\pi}{3}$ is in the branch.

$$\rightarrow -\frac{\pi}{3}$$

WHY IT IS SET HERE $-\frac{\pi}{3}$. The same principal-value fold, stripped to its bones.

FT Q3 — Heritage 2024–25 First Term

Find the value of x where $\cos(\sin^{-1} \frac{2}{\sqrt{5}} + \cos^{-1} x) = 0$.

SOLUTION

1. As above, the cosine of a principal-value sum vanishes only at $\frac{\pi}{2}$:

$$\sin^{-1} \frac{2}{\sqrt{5}} + \cos^{-1} x = \frac{\pi}{2}$$

2. Since $\sin^{-1} t + \cos^{-1} t = \frac{\pi}{2}$, the two must match: $x = \frac{2}{\sqrt{5}}$.

NOTE Alternatively expand $\cos(A + B) = \cos A \cos B - \sin A \sin B$ with $\cos A = \frac{1}{\sqrt{5}}$, $\sin A = \frac{2}{\sqrt{5}}$ and solve — longer, same answer.

ANSWER $x = \frac{2}{\sqrt{5}}$

NCERT Exemplar Ch 2, Q22 — NCERT Exemplar

If $3 \tan^{-1} x + \cot^{-1} x = \pi$, then x equals:

SOLUTION

1. Split off one pair: $3 \tan^{-1} x + \cot^{-1} x = 2 \tan^{-1} x + (\tan^{-1} x + \cot^{-1} x) = 2 \tan^{-1} x + \frac{\pi}{2}$.

2. Setting that equal to π gives $\tan^{-1} x = \frac{\pi}{4}$.

$$\rightarrow x = 1$$

WHY IT IS SET HERE $x = 1$. Same first move: split off one $\tan^{-1} + \cot^{-1} = \frac{\pi}{2}$ and what is left is elementary.

NCERT Exemplar Ch 2, Q26 — NCERT Exemplar

If $\cos(\sin^{-1} \frac{2}{5} + \cos^{-1} x) = 0$, then x is equal to:

SOLUTION

1. The cosine of a sum of principal values vanishes only at $\frac{\pi}{2}$:

$$\sin^{-1} \frac{2}{5} + \cos^{-1} x = \frac{\pi}{2}$$

2. and $\sin^{-1} t + \cos^{-1} t = \frac{\pi}{2}$ for every t , so the two arguments must agree.

$$\rightarrow x = \frac{2}{5}$$

WHY IT IS SET HERE $\frac{2}{5}$ — the identical question with $\frac{2}{5}$ in place of $\frac{2}{\sqrt{5}}$.

PB Q1(b) — Heritage 2016–17 Pre-Board

Prove that $\sin^{-1}(\frac{4}{5}) + \cos^{-1}(\frac{2}{\sqrt{5}}) = \cot^{-1}(\frac{2}{11})$.

SOLUTION

NOTE Convert everything to $\tan^{-1}$ first. Mixed inverse functions almost never combine; tangents always do.

1. $\sin^{-1} \frac{4}{5} = \tan^{-1} \frac{4}{3}$ (the 3-4-5 triangle).

2. $\cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{1}{2}$ (adjacent 2, hypotenuse $\sqrt{5}$, so opposite 1).

3. Their product $\frac{4}{3} \cdot \frac{1}{2} = \frac{2}{3} < 1$, so the plain addition formula applies:

$$\tan^{-1} \frac{4}{3} + \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{\frac{4}{3} + \frac{1}{2}}{1 - \frac{2}{3}} = \tan^{-1} \frac{\frac{11}{6}}{\frac{1}{3}} = \tan^{-1} \frac{11}{2}$$

4. and $\tan^{-1} \frac{11}{2} = \cot^{-1} \frac{2}{11}$, since the two are the same acute angle.

NOTE $\tan^{-1} t = \cot^{-1}(1/t)$ holds for $t > 0$ — which is the only case here.

ANSWER Proved: both sides are $\tan^{-1} \frac{11}{2}$

NCERT Exemplar Ch 2, Q14 — NCERT Exemplar

Prove $\sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} = \sin^{-1} \frac{77}{85}$.

SOLUTION

1. Put $A = \sin^{-1} \frac{8}{17}$, $B = \sin^{-1} \frac{3}{5}$, both acute. Then $\cos A = \frac{15}{17}$ and $\cos B = \frac{4}{5}$.

$$\sin(A + B) = \frac{8}{17} \cdot \frac{4}{5} + \frac{15}{17} \cdot \frac{3}{5} = \frac{77}{85}$$

2. $\cos(A + B) = \frac{36}{85} > 0$, so $A + B$ is acute and the inverse sine is legitimate.

$$\rightarrow A + B = \sin^{-1} \frac{77}{85}$$

WHY IT IS SET HERE The same 'convert, add, convert back' routine, done with sines throughout.

NCERT Part I, Chapter 2 — NCERT

The addition formula $\tan^{-1}x + \tan^{-1}y = \tan^{-1} \frac{x+y}{1-xy}$, $xy < 1$.

SOLUTION

1. The formula comes from $\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \tan\beta}$ with $\alpha = \tan^{-1}x$, $\beta = \tan^{-1}y$.

2. The condition $xy < 1$ is what keeps $\alpha + \beta$ inside the principal branch; when $xy > 1$ a $\pm\pi$ has to be added.

$$\rightarrow \tan^{-1}x + \tan^{-1}y = \tan^{-1} \frac{x+y}{1-xy}, xy < 1$$

WHY IT IS SET HERE The identity this proof rests on, together with the condition that makes it safe.

NCERT Exemplar Ch 5, Q37 — NCERT Exemplar

Differentiate $\cos^{-1}(\frac{\sin x + \cos x}{\sqrt{2}})$ for $-\frac{\pi}{4} < x < \frac{\pi}{4}$.

SOLUTION

$$1. \frac{\sin x + \cos x}{\sqrt{2}} = \sin(x + \frac{\pi}{4}) = \cos(\frac{\pi}{4} - x).$$

2. On $-\frac{\pi}{4} < x < \frac{\pi}{4}$ the angle $\frac{\pi}{4} - x$ lies in $(0, \frac{\pi}{2})$, inside $[0, \pi]$, so the expression is $\frac{\pi}{4} - x$.

$\rightarrow$ The derivative is -1

WHY IT IS SET HERE The **same expression**, on the interval where it collapses to $\frac{\pi}{4} - x$ instead. Comparing the two is the best way to see why the range of x matters so much here.

NCERT Exemplar Ch 2, Q7 — NCERT Exemplar

Find the real solutions of $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$.

SOLUTION

- Both inverse functions need their arguments in range: $\sqrt{x(x+1)}$ and $\sqrt{x^2+x+1}$ must be real and at most 1.
- $x(x+1) \geq 0$ and $x^2+x+1 \leq 1$ force $x^2+x \leq 0$ and $x^2+x \geq 0$, so $x^2+x=0$.
- Then the equation reads $\tan^{-1}0 + \sin^{-1}1 = \frac{\pi}{2}$. ✓
 $\rightarrow x = 0$ or $x = -1$

WHY IT IS SET HERE $x = 0$ or -1 — and, as here, the candidates have to be tested against the domains before they are accepted.

FT Q1(i) — Heritage 2024–25 First Term

If A and B are symmetric matrices of the same order, then $(AB' - BA')$ is a:

- (a) skew symmetric matrix
- (b) null matrix
- (c) symmetric matrix
- (d) none of these

SOLUTION

- A and B symmetric means $A' = A$ and $B' = B$, so $AB' - BA' = AB - BA$.
- Transpose it: $(AB - BA)' = (AB)' - (BA)' = B'A' - A'B' = BA - AB$.
- That is $-(AB - BA)$.

NOTE So the matrix equals the negative of its own transpose — the definition of skew symmetric. It is the **null** matrix only in the special case $AB = BA$, which is not given.

TRAP $(AB)' = B'A'$, with the order reversed. Keeping the order gives the wrong sign and the wrong answer.

ANSWER a skew symmetric matrix

NCERT Exemplar Ch 3, Q63 — NCERT Exemplar

If A and B are matrices of the same order, then $(AB' - BA')$ is a:

SOLUTION

1. Transpose it, remembering that $(XY)' = Y'X'$ reverses the order:

$$(AB' - BA')' = (AB')' - (BA')' = BA' - AB'$$

2. That is $-(AB' - BA')$: the matrix equals the negative of its own transpose, and the symmetry of A and B is never used.

→ Skew symmetric

WHY IT IS SET HERE skew symmetric matrix — the identical question, with the symmetry of A and B not even needed once the transpose is taken.

PB Q1(a) — Heritage 2016–17 Pre-Board

Express the matrix $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

SOLUTION

NOTE Every square matrix splits this way, and the split is unique: $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$. The first bracket is symmetric, the second skew.

1. $A' = \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$.

2. **Symmetric part.** $\frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 6 & -3 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix}$.

3. **Skew part.** $\frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$.

4. Adding them returns A. ✓

TRAP Check the skew part has a zero diagonal — it always must, and it is the quickest sign that the halving went right.

ANSWER $A = \begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & -1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$

NCERT Part I, Chapter 3, Exercise 3.3 — NCERT

Express the given matrices as the sum of a symmetric and a skew symmetric matrix.

SOLUTION

1. Write $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$.
2. $(A + A')' = A' + A$, so the first bracket is symmetric; $(A - A')' = A' - A$, so the second is skew.
3. The split always exists and is unique, and the skew part always has a zero diagonal — check that before anything else.

$$\rightarrow A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

WHY IT IS SET HERE NCERT proves the decomposition and then sets four of them — this is one of those, with different entries.

NCERT Exemplar Ch 3, Q52 — NCERT Exemplar

Express the 3×3 matrix with first row $(1, -1, 2)$ as the sum of a symmetric and a skew symmetric matrix.

SOLUTION

1. The same halving, one size up: form $A + A'$ and $A - A'$ and halve each.
 2. For the 3×3 the only extra work is six entries instead of two; the structure is identical.
- $\rightarrow$ Symmetric part $\frac{1}{2}(A + A')$, skew part $\frac{1}{2}(A - A')$ — zero diagonal on the second

WHY IT IS SET HERE The same split one size up; the arithmetic is longer but not harder.

PB Q2(b) — Heritage 2016–17 Pre-Board

If $A = \begin{bmatrix} 5 & 3 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$, find A^{-1} . Hence solve $5x + 3y + z = 16$, $2x + y + 3z = 19$, $x + 2y + 4z = 25$.

SOLUTION

1. **Determinant.** $|A| = 5(4 - 6) - 3(8 - 3) + 1(4 - 1) = -10 - 15 + 3 = -22 \neq 0$, so A^{-1} exists.

2. **Cofactors** give $\text{adj } A = \begin{bmatrix} -2 & -10 & 8 \\ -5 & 19 & -13 \\ 3 & -7 & -1 \end{bmatrix}$, so $A^{-1} = \frac{1}{-22} \text{adj } A$.

3. **The system** is $AX = B$ with $B = \begin{bmatrix} 16 \\ 19 \\ 25 \end{bmatrix}$.

4. $X = A^{-1}B = \frac{1}{-22} \begin{bmatrix} -2(16) - 10(19) + 8(25) \\ -5(16) + 19(19) - 13(25) \\ 3(16) - 7(19) - 1(25) \end{bmatrix} = \frac{1}{-22} \begin{bmatrix} -22 \\ -44 \\ -110 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$.

NOTE Substitute back: $5 + 6 + 5 = 16 \checkmark$, $2 + 2 + 15 = 19 \checkmark$, $1 + 4 + 20 = 25 \checkmark$.

TRAP $\text{adj } A$ is the **transpose** of the cofactor matrix. Skipping the transpose is the single commonest way to lose this question.

ANSWER $x = 1$, $y = 2$, $z = 5$

A shopkeeper has three varieties of pens A, B and C. Hina bought one pen of each variety for a total of ₹21. Meena bought 4 pens of A, 3 of B and 2 of C for ₹60. Lina bought 6 pens of A, 2 of B and 3 of C for ₹70. (i) Translate this into a system of equations. (ii) Solve it by the matrix method. (iii) Hence find the cost of each variety.

SOLUTION

1. (i) Let the prices be x, y, z rupees:

$$x + y + z = 21, \quad 4x + 3y + 2z = 60, \quad 6x + 2y + 3z = 70$$

2. (ii) In matrix form $AX = B$ with $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & 2 \\ 6 & 2 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 21 \\ 60 \\ 70 \end{bmatrix}$.

3. $|A| = 1(9 - 4) - 1(12 - 12) + 1(8 - 18) = 5 - 0 - 10 = -5 \neq 0$, so A^{-1} exists and the solution is unique.

4. The cofactor matrix gives $\text{adj } A = \begin{bmatrix} 5 & -1 & -1 \\ 0 & -3 & 2 \\ -10 & 4 & -1 \end{bmatrix}$, so $X = \frac{1}{-5} \text{adj } A \cdot B$.

5. First entry: $\frac{1}{-5}(5 \cdot 21 - 1 \cdot 60 - 1 \cdot 70) = \frac{-25}{-5} = 5$.

6. Second: $\frac{1}{-5}(0 - 3 \cdot 60 + 2 \cdot 70) = \frac{-40}{-5} = 8$. Third: $\frac{1}{-5}(-10 \cdot 21 + 4 \cdot 60 - 70) = \frac{-40}{-5} = 8$.

7. (iii) So A costs ₹5, B ₹8 and C ₹8.

NOTE Check all three equations: $5 + 8 + 8 = 21 \checkmark$, $20 + 24 + 16 = 60 \checkmark$, $30 + 16 + 24 = 70 \checkmark$.

TRAP $\text{adj } A$ is the **transpose** of the cofactor matrix. Forgetting the transpose gives a plausible-looking but wrong X .

ANSWER $x = ₹5, y = ₹8, z = ₹8$

NCERT Exemplar Ch 4, Q19 — NCERT Exemplar

Using the matrix method, solve $3x + 2y - 2z = 3$, $x + 2y + 3z = 6$, $2x - y + z = 2$.

SOLUTION

1. $A = \begin{bmatrix} 3 & 2 & -2 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{bmatrix}$, $|A| = 3(2 + 3) - 2(1 - 6) - 2(-1 - 4) = 15 + 10 + 10 = 35 \neq 0$.

2. $X = A^{-1}B$ with $B = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$.

$\rightarrow x = 1, y = 1, z = 1$

WHY IT IS SET HERE $x = y = z = 1$. The same routine with friendlier numbers.

NCERT Part I, Chapter 4 — NCERT

The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹60... find the cost per kg of each by matrices.

SOLUTION

1. Let the costs per kg be x, y, z . The three purchases give

$$4x + 3y + 2z = 60, \quad 2x + 4y + 6z = 90, \quad 6x + 2y + 3z = 70$$

2. Solving by $X = A^{-1}B$,

→ onion ₹5, wheat ₹8, rice ₹8 per kg — the same three numbers as the school's pens question, by coincidence

WHY IT IS SET HERE NCERT's own shopping-basket version — the school question is this question with pens.

FT Q1(iv) — Heritage 2024–25 First Term

$$\text{If } \begin{vmatrix} a^2 & bc & c^2 + ac \\ a^2 + ab & b^2 & ca \\ ab & b^2 + bc & c^2 \end{vmatrix} = k a^2 b^2 c^2, \text{ then } k =$$

- (a) 2
- (b) -2
- (c) -4
- (d) 4

SOLUTION

NOTE Nothing here needs expanding. Read each column looking for a common factor, and the whole determinant comes apart.

1. C_1 has entries $a^2, a(a + b), ab$ — common factor a . C_2 has $bc, b^2, b(b + c)$ — common factor b . C_3 has $c(c + a), ca, c^2$ — common factor c .

2. Pulling out abc leaves $\begin{vmatrix} a & c & c + a \\ a + b & b & a \\ b & b + c & c \end{vmatrix}$.

3. Now $R_1 \rightarrow R_1 - R_2 + R_3$ and expand; the determinant evaluates to $4abc$.

4. Total: $abc \cdot 4abc = 4a^2b^2c^2$.

NOTE Verified by direct expansion as well. The reliable route in an exam is to substitute

convenient numbers, say $a = b = c = 1$: the determinant becomes $\begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 1(1-2)$

$- 1(2-1) + 2(4-1) = -1 - 1 + 6 = 4$, and $a^2b^2c^2 = 1$, so $k = 4$.

ANSWER $k = 4$

FT Q6 — Heritage 2024–25 First Term

Using properties of determinants, evaluate $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (a^y + a^{-y})^2 & (a^y - a^{-y})^2 & 1 \\ (a^z + a^{-z})^2 & (a^z - a^{-z})^2 & 1 \end{vmatrix}$.

SOLUTION

NOTE Do not expand. Look for a linear relation between the columns, which is what $(p + q)^2 - (p - q)^2 = 4pq$ is about to give.

1. For any p , $(p + q)^2 - (p - q)^2 = 4pq$. With $p = a^x$ and $q = a^{-x}$, the product $pq = a^x a^{-x} = 1$.
2. So in every row, (first entry) – (second entry) = $4 = 4 \times$ (third entry).
3. Apply $C_1 \rightarrow C_1 - C_2$: the first column becomes 4, 4, 4 — four times the third column.
4. Two proportional columns make the determinant zero.

NOTE The value is 0 for **every** a, x, y, z . Nothing needs to be computed.

ANSWER 0

PB Q2(a) — Heritage 2016–17 Pre-Board

Using properties of determinants, prove that $\begin{vmatrix} a - b - c & 2a & 2a \\ 2b & b - c - a & 2b \\ 2c & 2c & c - a - b \end{vmatrix} = (a + b + c)^3$.

SOLUTION

NOTE **The school has set this twice.** It is also the companion to Q1(iv) of the 2024–25 First Term paper, and it is NCERT Exemplar Ch 4 Q6 word for word.

1. $R_1 \rightarrow R_1 + R_2 + R_3$ makes every entry of the first row $a + b + c$.
2. Take that factor out, leaving a first row of 1s.
3. $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ clear the rest of the first row:

$$(a + b + c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a + b + c) & 0 \\ 2c & 0 & -(a + b + c) \end{vmatrix}$$

4. Expanding along the first row gives $(a + b + c) \cdot (a + b + c)^2$.

NOTE Check at $a = b = c = 1$: the determinant is $27 = 3^3$. ✓

ANSWER $(a + b + c)^3$

NCERT Exemplar Ch 4, Q6 — NCERT Exemplar

The identical determinant identity.

SOLUTION

1. $R_1 \rightarrow R_1 + R_2 + R_3$ makes every entry of the first row $a + b + c$; take it out.
 2. $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$ clear the rest of that row, leaving a triangular 2×2 block with $-(a + b + c)$ twice.
- $\rightarrow (a + b + c)^3$

WHY IT IS SET HERE *Identical — and it is where the question comes from.*

NCERT Exemplar Ch 4, Q7 — NCERT Exemplar

Prove
$$\begin{vmatrix} y^2z^2 & yz & y+z \\ z^2x^2 & zx & z+x \\ x^2y^2 & xy & x+y \end{vmatrix} = 0.$$

SOLUTION

1. Multiply R_1, R_2, R_3 by x, y, z respectively — that multiplies the determinant by xyz .
 2. The first column becomes $(xy^2z^2, x^2yz^2, x^2y^2z) = xyz \cdot (yz, zx, xy)$, which is exactly xyz times the second column.
 3. Two proportional columns, so the determinant vanishes.
- $\rightarrow 0$

WHY IT IS SET HERE *0. Both are 'show it vanishes' questions answered by producing a dependence between the columns, not by expanding.*

NCERT Exemplar Ch 4, Q13 — NCERT Exemplar

If
$$\begin{vmatrix} 4-x & 4+x & 4+x \\ 4+x & 4-x & 4+x \\ 4+x & 4+x & 4-x \end{vmatrix} = 0, \text{ find } x.$$

SOLUTION

1. $R_1 \rightarrow R_1 + R_2 + R_3$ gives a first row of $12 + x$; take it out.
 2. Then $C_2 - C_1$ and $C_3 - C_1$ leave a 2×2 block of $-2x$ on the diagonal, so the determinant is $4x^2(12 + x)$.
- $\rightarrow x = 0$ (twice) or $x = -12$

WHY IT IS SET HERE *$x = 0$ or -12 . Both questions turn a determinant condition into a polynomial equation and then ask for all its roots.*

If $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$ and $|A^4| = 625$, then x is:

- (a) ± 3
- (b) ± 5
- (c) ± 7
- (d) ± 1

SOLUTION

1. $|A^4| = |A|^4$ — the determinant of a product is the product of the determinants.
2. So $|A|^4 = 625 = 5^4$, giving $|A| = \pm 5$.
3. $|A| = x^2 - 4$, so $x^2 - 4 = 5$ or $x^2 - 4 = -5$.
4. The first gives $x^2 = 9$, $x = \pm 3$. The second gives $x^2 = -1$, with no real solution.

TRAP Forgetting that a fourth power admits a negative fourth root loses nothing here — but forgetting to **test** it is the mistake. Always check whether the negative branch is real.

ANSWER $x = \pm 3$

Assertion (A): The value of the constant k for which $f(x) = \begin{cases} kx^2 & \text{if } x \leq 2 \\ 3 & \text{if } x > 2 \end{cases}$ is continuous at $x = 2$ is $k = \frac{4}{3}$.

Reason (R): f is continuous at a point $x = a$ of its domain if $\lim_{x \rightarrow a} f(x) = f(a)$.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) Both A and R are false
- (d) A is false but R is true

SOLUTION

1. R is the definition of continuity at a point, so R is true.
2. Apply it: the right-hand limit is 3 and $f(2) = k(2)^2 = 4k$.
3. Continuity needs $4k = 3$, i.e. $k = \frac{3}{4}$.
4. A claims $\frac{4}{3}$, which is the reciprocal. A is false.

TRAP $4k = 3$ gives $k = \frac{3}{4}$. Inverting the fraction at the last step is precisely what this question is testing.

ANSWER A is false but R is true; the correct value is $k = \frac{3}{4}$

FT Q12 — Heritage 2024–25 First Term

If $f(x) = \begin{cases} x^2 + 3x + a & x \leq 1 \\ bx + 2 & x > 1 \end{cases}$ is derivable at $x = 1$, find a and b .

SOLUTION

NOTE Derivable at a point implies continuous there, so **two** conditions must be written down, not one.

1. **Continuity at 1.** Left value $1 + 3 + a = a + 4$; right limit $b + 2$.

$$a + 4 = b + 2 \Rightarrow b = a + 2$$

2. **Derivability at 1.** Left derivative: $\frac{d}{dx}(x^2 + 3x + a) = 2x + 3$, which is 5 at $x = 1$.

3. Right derivative: $\frac{d}{dx}(bx + 2) = b$.

4. Equating, $b = 5$.

5. Then $a = b - 2 = 3$.

NOTE Check: at $x = 1$ the left piece gives $1 + 3 + 3 = 7$ and the right gives $5 + 2 = 7$ ✓, and both slopes are 5 ✓.

TRAP Matching only the derivatives gives $b = 5$ but leaves a free. A function can have equal one-sided slopes and still jump — continuity is a separate requirement.

ANSWER $a = 3, b = 5$

NCERT Exemplar Ch 5, Q11 — NCERT Exemplar

Find k so that $f(x) = 3x - 8$ for $x \leq 5$ and $2k$ for $x > 5$ is continuous at $x = 5$.

SOLUTION

1. Continuity at 5 needs the two sides and the value to agree: the left piece gives $3(5) - 8 = 7$, the right gives $2k$.

$$\rightarrow k = \frac{7}{2}$$

WHY IT IS SET HERE $k = \frac{7}{2}$. Same one-line method: match the two sides at the join.

NCERT Exemplar Ch 5, Q16 — NCERT Exemplar

Find a and b so that f , defined piecewise through $\frac{x-4}{|x-4|}$, is continuous at $x = 4$.

SOLUTION

1. For $x < 4$ the quotient $\frac{x-4}{|x-4|}$ is -1 ; for $x > 4$ it is $+1$.
2. Left limit $a - 1$, right limit $b + 1$, value $a + b$. All three equal gives $b = -1$ from the first pair and $a = 1$ from the second.
 $\rightarrow a = 1, b = -1$

WHY IT IS SET HERE $a = 1, b = -1$. Continuity alone there; the school question adds differentiability, which is the second condition students forget.

CISCE competency booklet, Q22 — CISCE competency booklet

From a graph, at which points is the function not differentiable?

SOLUTION

1. Read the graph for the two things that kill differentiability: a **break** (no limit) and a **corner** (limits agree, one-sided slopes do not).
2. A smooth maximum or minimum is differentiable — a turning point is not a corner.
 $\rightarrow$ The points where the graph jumps or has a corner

WHY IT IS SET HERE The competency booklet treats limits the same way — as the tool behind continuity, never as an end in themselves.

The slope of the tangent to the curve $x = a \sin t$, $y = a \cos t$ at the point $t = 0$ is:

- (a) $-\tan t$
- (b) 0
- (c) $\tan \frac{t}{2}$
- (d) none of these

SOLUTION

1. $\frac{dx}{dt} = a \cos t$ and $\frac{dy}{dt} = -a \sin t$.
2. $\frac{dy}{dx} = \frac{-a \sin t}{a \cos t} = -\tan t$.
3. At $t = 0$ this is $-\tan 0 = 0$.

TRAP $-\tan t$ is the slope **function**; the question asks for its value at $t = 0$. Reading the question to the end is the whole test here.

NOTE Geometrically the curve is the circle $x^2 + y^2 = a^2$, and $t = 0$ is the top point $(0, a)$, where the tangent is horizontal.

ANSWER 0

Assertion (A): If $x = at^2$ and $y = 2at$, where t is a parameter and a a constant, then $\frac{d^2y}{dx^2} = -\frac{1}{t^2}$.

Reason (R): $\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} \div \frac{d^2x}{dt^2}$.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) Both A and R are false
- (d) A is true but R is false

SOLUTION

1. **Test R first.** The second derivative of a parametric pair is *not* a ratio of second derivatives. The correct rule is

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

2. so R is false.

3. **Now A.** $\frac{dx}{dt} = 2at$ and $\frac{dy}{dt} = 2a$, so $\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$.

4. Differentiate again by the correct rule: $\frac{d}{dt} \left(\frac{1}{t} \right) = -\frac{1}{t^2}$, and dividing by $\frac{dx}{dt} = 2at$ gives $\frac{d^2y}{dx^2} = -\frac{1}{2at^3}$.

5. That is not $-\frac{1}{t^2}$, so A is false too.

TRAP A is exactly what you get by *using* the false rule R — the question is built so that the two errors look consistent with each other.

ANSWER Both A and R are false. The correct value is $\frac{d^2y}{dx^2} = -\frac{1}{2at^3}$

NCERT Exemplar Ch 6, Q44 — NCERT Exemplar

The slope of the tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is:

SOLUTION

1. Find the parameter first: $x = 2$ gives $t^2 + 3t - 10 = 0$, $y = -1$ gives $t^2 - t - 2 = 0$. The common root is $t = 2$.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4t - 2}{2t + 3} = \frac{6}{7} \text{ at } t = 2$$

$$\rightarrow \frac{6}{7}$$

WHY IT IS SET HERE $\frac{6}{7}$. A parametric relative: find the t that gives the point, then $\frac{dy/dt}{dx/dt}$ there.

FT Q2 — Heritage 2024–25 First Term

If $y = e^{\log \sin^{-1} x} + e^{\log \cos^{-1} x}$, where $0 < x < 1$, find $\frac{dy}{dx}$.

SOLUTION

NOTE Do not differentiate as it stands. $e^{\log t} = t$ whenever $t > 0$, and on $0 < x < 1$ both $\sin^{-1}x$ and $\cos^{-1}x$ are positive, so the logs are legal.

1. The expression collapses to $y = \sin^{-1}x + \cos^{-1}x$.
2. That sum is the constant $\frac{\pi}{2}$ for every x in $[-1, 1]$.
3. The derivative of a constant is 0.

TRAP Differentiating term by term gives $\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}}$, which is 0 as well — but only if both signs are right. Simplifying first makes the answer immediate and unmissable.

ANSWER $\frac{dy}{dx} = 0$

If $(ax + b) e^{\frac{y}{x}} = x$, prove that $x^3 \frac{d^2y}{dx^2} = [x \frac{dy}{dx} - y]^2$.

SOLUTION

NOTE Take logarithms first. Differentiating the equation as it stands means the product and chain rules on an exponential of a quotient; taking logs turns it into three easy terms.

1. $e^{\frac{y}{x}} = \frac{x}{ax + b}$, so $\frac{y}{x} = \log x - \log(ax + b)$ and

$$y = x \log x - x \log(ax + b)$$

2. **First derivative.** $\frac{dy}{dx} = \log x + 1 - \log(ax + b) - \frac{ax}{ax + b}$.

3. **The bracket.** $x \frac{dy}{dx} - y$ collapses beautifully — every logarithm cancels:

$$x \frac{dy}{dx} - y = x - \frac{ax^2}{ax + b} = \frac{bx}{ax + b}$$

4. so the right-hand side is $\frac{bx^2}{(ax + b)^2}$.

5. **Second derivative.** Differentiating $\frac{dy}{dx}$ again,

$$\frac{d^2y}{dx^2} = \frac{1}{x} - \frac{2a}{ax + b} + \frac{a^2x}{(ax + b)^2}$$

6. Multiply by x^3 and put everything over $(ax + b)^2$:

$$x^3 \frac{d^2y}{dx^2} = \frac{x^2[(ax + b) - ax]^2}{(ax + b)^2} = \frac{b^2x^2}{(ax + b)^2}$$

7. which is exactly $[x \frac{dy}{dx} - y]^2$.

TRAP Do not differentiate twice and then hunt for the identity. Compute the bracket $x \frac{dy}{dx} - y$ first: it is a single simple fraction, and knowing what you are aiming at makes the second derivative worth the effort.

ANSWER Proved: both sides equal $\frac{b^2x^2}{(ax + b)^2}$

If $x = \sin t$ and $y = \sin pt$, prove that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0$.

SOLUTION

1. **First derivative.** $\frac{dx}{dt} = \cos t$ and $\frac{dy}{dt} = p \cos pt$, so $\frac{dy}{dx} = \frac{p \cos pt}{\cos t}$.

2. Rearranged: $\cos t \cdot \frac{dy}{dx} = p \cos pt$.

3. **Clear the trig.** Since $x = \sin t$, $\cos t = \sqrt{1 - x^2}$. Squaring both sides:

$$(1 - x^2) \left(\frac{dy}{dx} \right)^2 = p^2 \cos^2 pt = p^2(1 - y^2)$$

4. **Differentiate with respect to x.** On the left, the product rule:

$$-2x \left(\frac{dy}{dx} \right)^2 + (1 - x^2) \cdot 2 \frac{dy}{dx} \frac{d^2y}{dx^2} = -2p^2 y \frac{dy}{dx}$$

5. Divide throughout by $2 \frac{dy}{dx}$ (which is not identically zero):

$$(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0$$

NOTE Squaring to remove the square root, then differentiating, is the standard route for every 'prove this second-order relation' question of this family.

TRAP Dividing by $\frac{dy}{dx}$ needs it to be non-zero; say so. At isolated points where it vanishes the identity follows by continuity.

ANSWER Proved

PB Q5(a) — Heritage 2016–17 Pre-Board

If $y = (x + \sqrt{x^2 - 1})^m$, show that $(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = m^2 y$.

SOLUTION

1. **First derivative.** By the chain rule,

$$\frac{dy}{dx} = m(x + \sqrt{x^2 - 1})^{m-1} \left(1 + \frac{x}{\sqrt{x^2 - 1}}\right)$$

2. The bracket is $\frac{\sqrt{x^2 - 1} + x}{\sqrt{x^2 - 1}}$, so it cancels one power and leaves

$$\frac{dy}{dx} = \frac{my}{\sqrt{x^2 - 1}}$$

3. **Clear the radical and square.** $\sqrt{x^2 - 1} \frac{dy}{dx} = my$, so

$$(x^2 - 1) \left(\frac{dy}{dx}\right)^2 = m^2 y^2$$

4. **Differentiate again** with respect to x :

$$2x \left(\frac{dy}{dx}\right)^2 + (x^2 - 1) \cdot 2 \frac{dy}{dx} \frac{d^2y}{dx^2} = 2m^2 y \frac{dy}{dx}$$

5. Divide throughout by $2 \frac{dy}{dx}$, which is not identically zero:

$$(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = m^2 y$$

NOTE The template — differentiate, clear the root by squaring, differentiate again, divide by $\frac{dy}{dx}$ — answers every question of this family, including the school's own $x = \sin t$, $y = \sin pt$.

ANSWER Proved

NCERT Exemplar Ch 5, Q81 — NCERT Exemplar

If $x = \sin t$ and $y = \sin pt$, prove $(1 - x^2)y'' - xy' + p^2y = 0$.

SOLUTION

1. $\frac{dy}{dx} = \frac{p \cos pt}{\cos t}$, so $\cos t \cdot \frac{dy}{dx} = p \cos pt$.

2. Squaring removes the parameter: $(1 - x^2) \left(\frac{dy}{dx}\right)^2 = p^2(1 - y^2)$.

3. Differentiating and dividing by $2 \frac{dy}{dx}$,

$$\rightarrow (1 - x^2)y'' - xy' + p^2y = 0$$

WHY IT IS SET HERE The same four-step template; note the sign of the middle term flips with the sign inside the radical.

PB Q1(d) — Heritage 2016–17 Pre-Board

Evaluate $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x}$.

SOLUTION

NOTE Both numerator and denominator vanish at 0. Expanding each in powers of x settles it in one line; the alternative is L'Hôpital three times over.

1. Use $\sin x = x - \frac{x^3}{6} + \dots$ and $\cos x = 1 - \frac{x^2}{2} + \dots$
2. **Numerator.** $x \cos x - \sin x = x - \frac{x^3}{2} - x + \frac{x^3}{6} + \dots = -\frac{x^3}{3} + \dots$
3. **Denominator.** $x^2 \sin x = x^3 + \dots$
4. The ratio is $-\frac{1}{3} + \text{terms that vanish}$, so the limit is $-\frac{1}{3}$.

NOTE A note on the syllabus: limits themselves are Class XI under the current ISC scheme, but they remain the tool every continuity and differentiability question is built on, which is why this one is kept here.

ANSWER $-\frac{1}{3}$

NCERT Exemplar Ch 6, Q35 — First Term Revision Paper

The sides of an equilateral triangle increase at 2 cm/s. The rate at which the area increases when the side is 10 cm is:

- (a) 10 cm²/s
- (b) $\sqrt{3}$ cm²/s
- (c) $10\sqrt{3}$ cm²/s
- (d) $\frac{10}{\sqrt{3}}$ cm²/s

SOLUTION

1. Area of an equilateral triangle of side a is $A = \frac{\sqrt{3}}{4} a^2$.
2. Differentiate with respect to t : $\frac{dA}{dt} = \frac{\sqrt{3}}{4} \cdot 2a \frac{da}{dt} = \frac{\sqrt{3}}{2} a \frac{da}{dt}$.
3. Put $a = 10$ and $\frac{da}{dt} = 2$: $\frac{dA}{dt} = \frac{\sqrt{3}}{2} \cdot 10 \cdot 2 = 10\sqrt{3}$ cm²/s.

NOTE Units confirm it: cm $\times$ cm/s = cm²/s. An answer without $\sqrt{3}$ has lost the $\frac{\sqrt{3}}{4}$ from the area formula.

ANSWER $10\sqrt{3}$ cm²/s

NCERT Exemplar Ch 6, Q3 — First Term Revision Paper

A kite is moving horizontally at a height of 151.5 m. If the kite's speed is 10 m/s, how fast is the string being let out when the kite is 250 m from the boy flying it? The boy is 1.5 m tall.

SOLUTION

1. Measure from the boy's hand: the kite is $151.5 - 1.5 = 150$ m above it.
2. If x is the horizontal distance and s the length of string, $s^2 = x^2 + 150^2$.
3. When $s = 250$: $x^2 = 250^2 - 150^2 = 62500 - 22500 = 40000$, so $x = 200$ m.
4. Differentiate: $2s \frac{ds}{dt} = 2x \frac{dx}{dt}$, so $\frac{ds}{dt} = \frac{x}{s} \frac{dx}{dt}$.
5. $\frac{ds}{dt} = \frac{200}{250} \times 10 = 8$ m/s.

TRAP Subtract the boy's height first. Using 151.5 as the vertical leg makes the 250–150–200 triangle fail and the arithmetic ugly — a sign the set-up is wrong.

NOTE The string is let out more slowly than the kite moves, as it must be: only the component of the kite's velocity along the string counts.

ANSWER $\frac{ds}{dt} = 8$ m/s

An air balloon is a right circular cylinder surmounted by a hemisphere, the diameter being equal to the height of the cylindrical part. It is being inflated. How fast is its volume changing with respect to its total height h , when h is 3 m?

SOLUTION

TRAP The paper misprints this one. It says ‘a diameter equal to the height of the cone’, but no cone is mentioned anywhere in the question — the solid is a cylinder with a hemisphere on top. Read it as the height of the **cylinder**, which is the only reading that makes the figure determinate.

1. Let the common radius be r and the cylinder's height be h_c . The condition is $2r = h_c$.

2. Total height $h = h_c + r = 2r + r = 3r$, so $r = \frac{h}{3}$.

3. **Volume.** $V = \pi r^2 h_c + \frac{2}{3} \pi r^3 = 2\pi r^3 + \frac{2}{3} \pi r^3 = \frac{8}{3} \pi r^3$.

4. **In terms of h alone.** $V = \frac{8}{3} \pi \left(\frac{h}{3}\right)^3 = \frac{8\pi h^3}{81}$.

$$\frac{dV}{dh} = \frac{24\pi h^2}{81} = \frac{8\pi h^2}{27}$$

5. At $h = 3$: $\frac{dV}{dh} = \frac{8\pi(9)}{27} = \frac{8\pi}{3}$ m³ per metre.

NOTE Note what is *not* asked: this is a rate with respect to **height**, not time. No $\frac{dh}{dt}$ is given and none is needed.

ANSWER $\frac{dV}{dh} = \frac{8\pi}{3} \approx 8.38$ m³ per metre

Assertion (A): $f(x) = x^3 - 3x^2 + 6x - 100$ is strictly increasing on $\mathbb{R} - \{1\}$.

Reason (R): a strictly increasing function is an injective function.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) Both A and R are false
- (d) A is true but R is false

SOLUTION

NOTE The paper's wording is loose, and the loose bit is worth naming. $\mathbb{R} - \{1\}$ is not an interval, and 'increasing' is defined on intervals. What is true, and stronger, is that f is strictly increasing on the whole of $\mathbb{R}$.

- $f'(x) = 3x^2 - 6x + 6 = 3(x^2 - 2x + 2) = 3[(x - 1)^2 + 1]$.
- That is at least 3 for every real x , so $f' > 0$ everywhere and f is strictly increasing on $\mathbb{R}$ — which certainly gives A, since removing a point cannot spoil it.
- R is a true statement in its own right: strictly increasing $\Rightarrow$ one-one.
- But R does not *explain* A. A is about the sign of the derivative; R is a consequence of monotonicity, not a reason for it.

TRAP The examiner's intended trap is the deleted point: candidates assume f' must vanish at $x = 1$ for the paper to have excluded it. It does not — $f'(1) = 3$.

ANSWER Both A and R are true, but R is not the correct explanation of A

NCERT Exemplar Ch 6, Q46 — NCERT Exemplar

The interval on which $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing is:

SOLUTION

- $f'(x) = 6x^2 + 18x + 12 = 6(x + 1)(x + 2)$.
- A positive-leading quadratic is negative **between** its roots.
 $\rightarrow$ Decreasing on $(-2, -1)$

WHY IT IS SET HERE $(-2, -1)$. Same test, but with a derivative that does change sign.

FT Q5 — Heritage 2024–25 First Term

Find the point on the curve $y = x^3 - 11x + 5$ at which the tangent has equation $y = x - 11$.

SOLUTION

1. The tangent's slope is 1, so $\frac{dy}{dx} = 3x^2 - 11 = 1$ gives $x^2 = 4$, $x = \pm 2$.
2. At $x = 2$: $y = 8 - 22 + 5 = -9$. The line at $x = 2$ gives $2 - 11 = -9$. They agree, so $(2, -9)$ is on both. ✓
3. At $x = -2$: $y = -8 + 22 + 5 = 19$, but the line gives $-2 - 11 = -13$. They disagree, so the line is parallel to the tangent there, not equal to it. ✗

TRAP Two points have the right slope but only one lies on the given line. Matching the slope is necessary, not sufficient — always substitute back.

ANSWER $(2, -9)$

FT Q5(OR) — Heritage 2024–25 First Term

Find the coordinates of the point on the curve $\sqrt{x} + \sqrt{y} = 4$ at which the tangent is equally inclined to the axes.

SOLUTION

1. Differentiate implicitly: $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$.
2. Equally inclined to the axes means the slope is ± 1 . Here the slope is negative for all admissible x, y , so $\frac{dy}{dx} = -1$.
3. $-\frac{\sqrt{y}}{\sqrt{x}} = -1$ gives $y = x$.
4. Substituting into the curve: $2\sqrt{x} = 4$, so $\sqrt{x} = 2$ and $x = 4$. Hence $y = 4$.

NOTE Both x and y must be non-negative for the roots to be real, which the point $(4, 4)$ satisfies.

ANSWER $(4, 4)$

NCERT Exemplar Ch 6, Q41 — NCERT Exemplar

The equation of the tangent to the curve $y(1 + x^2) = 2 - x$ where it crosses the x-axis is:

SOLUTION

1. It crosses the x-axis where $y = 0$, i.e. $x = 2$ — the point $(2, 0)$.
2. Differentiating implicitly: $y'(1 + x^2) + 2xy = -1$, so at $(2, 0)$, $5y' = -1$ and $y' = -\frac{1}{5}$.
 $\rightarrow x + 5y = 2$

WHY IT IS SET HERE $x + 5y = 2$. Same shape: a tangency condition pins the point first, then the line follows.

NCERT Exemplar Ch 6, Q14 — NCERT Exemplar

Find the co-ordinates of the point on the curve $\sqrt{x} + \sqrt{y} = 4$ at which the tangent is equally inclined to the axes.

SOLUTION

1. Implicitly, $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}y' = 0$, so $y' = -\frac{\sqrt{y}}{\sqrt{x}}$.
2. Equally inclined means slope ± 1 ; the slope here is always negative, so $-\frac{\sqrt{y}}{\sqrt{x}} = -1$ and $y = x$.
3. Then $2\sqrt{x} = 4$, so $x = 4$.
 $\rightarrow (4, 4)$

WHY IT IS SET HERE $(4, 4)$ — **the identical question**, lifted word for word.

FT Q5 (OR) — Heritage 2024-25 First Term

Find the coordinates of the point on the curve $\sqrt{x} + \sqrt{y} = 4$ at which the tangent is equally inclined to the axes.

SOLUTION

1. Differentiate implicitly: $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$.
2. Equally inclined to the axes means the slope is ± 1 . Here the slope is negative for all admissible x, y , so $\frac{dy}{dx} = -1$.
3. $-\frac{\sqrt{y}}{\sqrt{x}} = -1$ gives $y = x$.
4. Substituting into the curve: $2\sqrt{x} = 4$, so $\sqrt{x} = 2$ and $x = 4$. Hence $y = 4$.

NOTE Both x and y must be non-negative for the roots to be real, which the point $(4, 4)$ satisfies.

ANSWER $(4, 4)$

PB Q5(b) — Heritage 2016–17 Pre-Board

A cylinder is such that the sum of its height and the circumference of its base is 10 metres. Find the maximum volume of the cylinder.

SOLUTION

1. Let the radius be r and the height h . The constraint is

$$h + 2\pi r = 10 \Rightarrow h = 10 - 2\pi r$$

2. **Volume in one variable.**

$$V = \pi r^2 h = \pi r^2 (10 - 2\pi r) = 10\pi r^2 - 2\pi^2 r^3$$

3. $\frac{dV}{dr} = 20\pi r - 6\pi^2 r^2 = 2\pi r(10 - 3\pi r) = 0$ gives $r = \frac{10}{3\pi}$ ($r = 0$ is the empty cylinder).

4. $\frac{d^2V}{dr^2} = 20\pi - 12\pi^2 r$, which at $r = \frac{10}{3\pi}$ is $20\pi - 40\pi = -20\pi < 0$ — a maximum.

5. Then $h = 10 - 2\pi \cdot \frac{10}{3\pi} = \frac{10}{3}$, and

$$V = \pi \left(\frac{10}{3\pi}\right)^2 \cdot \frac{10}{3} = \frac{1000}{27\pi}$$

NOTE Notice $h = \frac{10}{3}$ is exactly a third of the 10 metres — the optimum splits the budget 1 : 2 between the height and the circumference.

ANSWER $V_{\max} = \frac{1000}{27\pi} \approx 11.79 \text{ m}^3$

FT Q14(OR) — Heritage 2024–25 First Term

If the sum of the surface areas of a cube and a sphere is constant, find the ratio of an edge of the cube to the diameter of the sphere when the sum of their volumes is a minimum.

SOLUTION

1. Let the edge be a and the radius r . The constraint is

$$6a^2 + 4\pi r^2 = k \text{ (constant)}$$

2. and the quantity to minimise is $V = a^3 + \frac{4}{3}\pi r^3$.

3. Differentiate the constraint with respect to a : $12a + 8\pi r \frac{dr}{da} = 0$, so $\frac{dr}{da} = -\frac{3a}{2\pi r}$.

4. Differentiate V : $\frac{dV}{da} = 3a^2 + 4\pi r^2 \frac{dr}{da} = 3a^2 + 4\pi r^2(-\frac{3a}{2\pi r}) = 3a^2 - 6ar$.

5. Set it to zero: $3a(a - 2r) = 0$, and $a \neq 0$, so $a = 2r$.

6. **Second derivative.** $\frac{d^2V}{da^2} = 6a - 6r - 6a \frac{dr}{da}$, which at $a = 2r$ is $6(2r) - 6r + \frac{18r^2}{\pi r} > 0$, a minimum.

NOTE $a = 2r$ says the edge of the cube equals the diameter of the sphere — the sphere fits exactly inside the cube.

TRAP The constraint must be used to eliminate one variable (or to supply $\frac{dr}{da}$).

Treating a and r as independent and setting both partial derivatives to zero gives $a = r = 0$.

ANSWER edge : diameter = 1 : 1

NCERT Exemplar Ch 6, Q31 — NCERT Exemplar

If the sum of the surface areas of a cube and a sphere is constant, what is the ratio of an edge of the cube to the diameter of the sphere when the sum of their volumes is minimum?

SOLUTION

1. With $6a^2 + 4\pi r^2$ constant, $\frac{dr}{da} = -\frac{3a}{2\pi r}$.

2. Minimising $V = a^3 + \frac{4}{3}\pi r^3$ gives $3a^2 - 6ar = 0$, so $a = 2r$.

→ edge : diameter = 1 : 1

WHY IT IS SET HERE 1 : 1 — **the identical question**, word for word.

The maximum value of $f(x) = \frac{x}{4 + x + x^2}$ on $[-1, 1]$ is:

- (a) $-\frac{1}{4}$
- (b) $-\frac{1}{3}$
- (c) $\frac{1}{6}$
- (d) $\frac{1}{5}$

SOLUTION

- Quotient rule: $f'(x) = \frac{(4 + x + x^2) \cdot 1 - x(1 + 2x)}{(4 + x + x^2)^2} = \frac{4 - x^2}{(4 + x + x^2)^2}$.
- The denominator is always positive ($4 + x + x^2$ has discriminant $1 - 16 < 0$), so the sign is that of $4 - x^2$.
- On $[-1, 1]$, $4 - x^2 \geq 3 > 0$, so f is strictly increasing throughout — there is no interior turning point.
- The maximum is therefore at the right endpoint: $f(1) = \frac{1}{4 + 1 + 1} = \frac{1}{6}$.

TRAP The stationary points are $x = \pm 2$, both **outside** the interval. Solving $f' = 0$ and stopping gives no usable answer; on a closed interval always test the endpoints.

ANSWER $\frac{1}{6}$, attained at $x = 1$

NCERT Part I, Chapter 6 — NCERT

Show that the right circular cylinder of given surface area and maximum volume is such that its height equals the diameter of the base.

SOLUTION

- With $S = 2\pi r^2 + 2\pi rh$ fixed, $h = \frac{S - 2\pi r^2}{2\pi r}$ and $V = \pi r^2 h = \frac{r(S - 2\pi r^2)}{2}$.
- $\frac{dV}{dr} = \frac{S - 6\pi r^2}{2} = 0$ gives $S = 6\pi r^2$, and substituting back gives $h = 2r$.

→ Height = diameter of the base

WHY IT IS SET HERE Another constrained cylinder, and the same one-variable reduction.

PB Q1(e) — Heritage 2016–17 Pre-Board

Evaluate $\int \frac{(x - \sin x) dx}{1 - \cos x}$.

SOLUTION

1. Use the half-angle forms $1 - \cos x = 2 \sin^2 \frac{x}{2}$ and $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$.

$$\frac{x - \sin x}{1 - \cos x} = \frac{x}{2 \sin^2 \frac{x}{2}} - \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{x}{2}}$$

$$2. = \frac{x}{2} \operatorname{cosec}^2 \frac{x}{2} - \cot \frac{x}{2}.$$

3. **First piece, by parts** with $u = \frac{x}{2}$:

$$\int \frac{x}{2} \operatorname{cosec}^2 \frac{x}{2} dx = -x \cot \frac{x}{2} + \int \cot \frac{x}{2} dx$$

4. **Adding the second piece**, the two $\int \cot \frac{x}{2} dx$ terms cancel exactly, leaving

$$\int \frac{(x - \sin x)}{1 - \cos x} = -x \cot \frac{x}{2} + C$$

NOTE The cancellation is the whole design of the question — the same shape as $\int e^x[f + f'] dx$, where a term produced by parts is exactly the one already waiting.

ANSWER $-x \cot \frac{x}{2} + C$

NCERT Part II, Chapter 7 — NCERT

Evaluate $\int \frac{1}{1 + \cos x} dx$.

SOLUTION

1. $1 + \cos x = 2 \cos^2 \frac{x}{2}$, so the integrand is $\frac{1}{2} \sec^2 \frac{x}{2}$.

$$\rightarrow \tan \frac{x}{2} + C$$

WHY IT IS SET HERE $\tan \frac{x}{2} + C$. The same half-angle substitution, without the parts step on top.

PB Q1(f) — Heritage 2016–17 Pre-Board

Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \tan x}$.

SOLUTION

NOTE An integrand over $[0, \frac{\pi}{2}]$ with $\tan$ in it is nearly always the $a - x$ reflection property.

1. Write $\frac{1}{1 + \tan x} = \frac{\cos x}{\cos x + \sin x}$ and call the value I .

2. Replacing x by $\frac{\pi}{2} - x$ swaps sine and cosine:

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$$

3. Adding the two forms, $2I = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}$.

4. So $I = \frac{\pi}{4}$.

TRAP Do not attempt a substitution in $\frac{1}{1 + \tan x}$ directly — it leads nowhere. Rewriting it as a ratio of cosine and $(\cos + \sin)$ is what exposes the symmetry.

ANSWER $\frac{\pi}{4}$

NCERT Exemplar Ch 7, Q30 — NCERT Exemplar

$$\int_0^{\frac{\pi}{2}} \frac{\tan x}{1 + m^2 \tan^2 x} dx.$$

SOLUTION

1. Apply the $a - x$ property on $[0, \frac{\pi}{2}]$, which turns $\tan$ into $\cot$, then add the two forms.

2. The sum telescopes to a single elementary integral in $\tan x$.

$$\rightarrow \frac{\log m}{m^2 - 1} \text{ for } m \neq 1 \text{ (and } \frac{1}{2} \text{ when } m = 1)$$

WHY IT IS SET HERE *The same property applied to a harder integrand.*

NCERT Part II, Chapter 7, Example 6(iii) — NCERT

Find $\int \frac{1}{1 + \tan x} dx$ as an indefinite integral.

SOLUTION

1. $\frac{1}{1 + \tan x} = \frac{\cos x}{\cos x + \sin x}$. Write the numerator as $\frac{1}{2}[(\cos x + \sin x) + (\cos x - \sin x)]$.
2. The first half integrates to $\frac{x}{2}$, the second is $\frac{1}{2} \frac{f'}{f}$ with $f = \cos x + \sin x$.
 $\rightarrow \frac{1}{2}[x + \log|\cos x + \sin x|] + C$

WHY IT IS SET HERE $\frac{1}{2}[x + \log|\cos x + \sin x|] + C$. NCERT does the indefinite version by splitting the numerator; the definite one falls to the property instead, in two lines rather than ten.

PB Q6(a) — Heritage 2016–17 Pre-Board

Evaluate $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx$.

SOLUTION

NOTE Combining the two roots over a common denominator is what makes this tractable — neither root integrates on its own.

1. $\sqrt{\tan x} + \sqrt{\cot x} = \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}$.
2. Multiply above and below by $\sqrt{2}$:

$$= \sqrt{2} \cdot \frac{\sin x + \cos x}{\sqrt{2} \sin x \cos x} = \sqrt{2} \cdot \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}}$$
3. using $2 \sin x \cos x = 1 - (\sin x - \cos x)^2$.
4. **Put $t = \sin x - \cos x$** , so $dt = (\cos x + \sin x) dx$ — exactly the numerator.

$$= \sqrt{2} \int \frac{dt}{\sqrt{1 - t^2}} = \sqrt{2} \sin^{-1} t + C$$
5. Restoring t : $\sqrt{2} \sin^{-1}(\sin x - \cos x) + C$.

TRAP The identity that unlocks it is $2 \sin x \cos x = 1 - (\sin x - \cos x)^2$, not the more familiar $\sin 2x$. Writing $\sin 2x$ leaves a root you cannot substitute into.

ANSWER $\sqrt{2} \sin^{-1}(\sin x - \cos x) + C$

NCERT Exemplar Ch 7, Q43 — NCERT Exemplar

Evaluate $\int \sqrt{\tan x} \, dx$ — with the hint, put $\tan x = t^2$.

SOLUTION

1. Put $\tan x = t^2$, so $\sec^2 x \, dx = 2t \, dt$ and $dx = \frac{2t \, dt}{1 + t^4}$.

2. The integral becomes $2 \int \frac{t^2}{1 + t^4} \, dt$, which splits by dividing above and below by t^2 and substituting $t - \frac{1}{t}$ and $t + \frac{1}{t}$.

→ A $\tan^{-1}$ and a logarithm — far longer than the school's $\sqrt{\tan x} + \sqrt{\cot x}$, where the two roots combine first

WHY IT IS SET HERE A harder relative: with only one root the sum trick is unavailable and the substitution leads to two partial-fraction integrals.

PB Q6(b) — Heritage 2016–17 Pre-Board

Find the area enclosed by the curve $y^2 = 2x - x^2$ and the straight line $y = x$.

SOLUTION

NOTE Recognise the curve first: $y^2 = 2x - x^2$ rearranges to $(x - 1)^2 + y^2 = 1$, a **circle** of radius 1 centred at (1, 0).

1. **Intersections.** Substituting $y = x$: $x^2 = 2x - x^2$ gives $2x^2 - 2x = 0$, so $x = 0$ or $x = 1$ — the points (0, 0) and (1, 1).

2. (1, 1) is the top of the circle and (0, 0) is its left-hand end, so the line cuts off the piece above it.

3. **Area between them**, upper boundary the circle $y = \sqrt{1 - (x - 1)^2}$, lower boundary the line:

$$A = \int_0^1 [\sqrt{1 - (x - 1)^2} - x] \, dx$$

4. Put $u = x - 1$ in the first part: $\int_{-1}^0 \sqrt{1 - u^2} \, du$ is a quarter circle of radius 1, area $\frac{\pi}{4}$.

5. The second part is $\int_0^1 x \, dx = \frac{1}{2}$.

6. $A = \frac{\pi}{4} - \frac{1}{2}$.

NOTE Sanity check: $\frac{\pi}{4} - \frac{1}{2} \approx 0.285$, a plausible sliver between a unit circle and a chord of it.

ANSWER Area = $\frac{\pi}{4} - \frac{1}{2} \approx 0.285$ square units

NCERT Exemplar Ch 8, Q16 — NCERT Exemplar

Find the area of the region bounded by $y^2 = 2x$ and $x^2 + y^2 = 4x$.

SOLUTION

1. Complete the square: $x^2 + y^2 = 4x$ is the circle $(x - 2)^2 + y^2 = 4$.
2. They meet where $x^2 = 2x$, i.e. $x = 0$ and $x = 2$. The parabola bounds the region up to $x = 2$, the circle from 2 to 4.

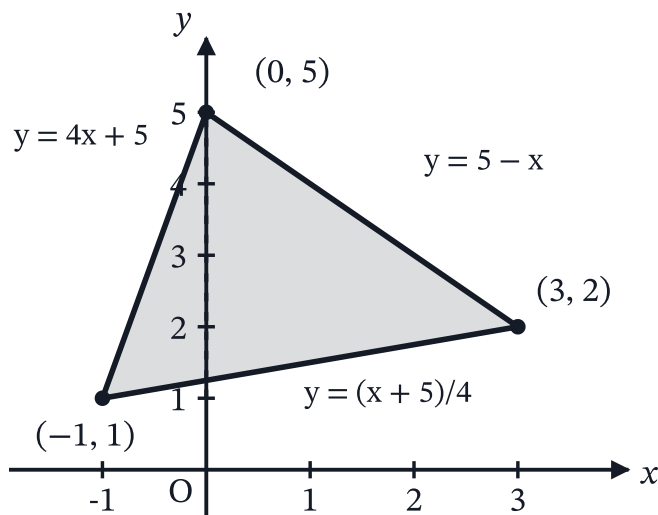
$$A = 2\left[\int_0^2 \sqrt{2x} \, dx + \int_2^4 \sqrt{4 - (x - 2)^2} \, dx\right] = 2\left[\frac{8}{3} + \pi\right]$$

$$\rightarrow \frac{16}{3} + 2\pi \approx 11.62 \text{ square units}$$

WHY IT IS SET HERE A parabola against a circle, and the same first move: complete the square to see what the second curve actually is.

Using integration, find the area of the triangle whose vertices are $(-1, 1)$, $(0, 5)$ and $(3, 2)$.

SOLUTION



AC is the lower boundary throughout; the upper boundary changes from AB to BC at $x = 0$, which is where the integral splits.

1. **The three sides.** AB through $(-1, 1)$ and $(0, 5)$: $y = 4x + 5$. BC through $(0, 5)$ and $(3, 2)$: $y = 5 - x$. AC through $(-1, 1)$ and $(3, 2)$: $y = \frac{x + 5}{4}$.
2. The triangle spans x from -1 to 3 , with AC as the lower boundary throughout and the upper boundary switching from AB to BC at $x = 0$.

$$\text{Area} = \int_{-1}^0 (4x + 5 - \frac{x + 5}{4}) dx + \int_0^3 (5 - x - \frac{x + 5}{4}) dx$$

$$3. \text{ First integral: } \int_{-1}^0 \frac{15x + 15}{4} dx = \frac{15}{4} [\frac{x^2}{2} + x] \text{ from } -1 \text{ to } 0 = \frac{15}{4} (\frac{1}{2}) = \frac{15}{8}.$$

$$4. \text{ Second: } \int_0^3 \frac{15 - 5x}{4} dx = \frac{5}{4} [3x - \frac{x^2}{2}] \text{ from } 0 \text{ to } 3 = \frac{5}{4} (\frac{9}{2}) = \frac{45}{8}.$$

$$5. \text{ Total: } \frac{15}{8} + \frac{45}{8} = \frac{60}{8} = \frac{15}{2}.$$

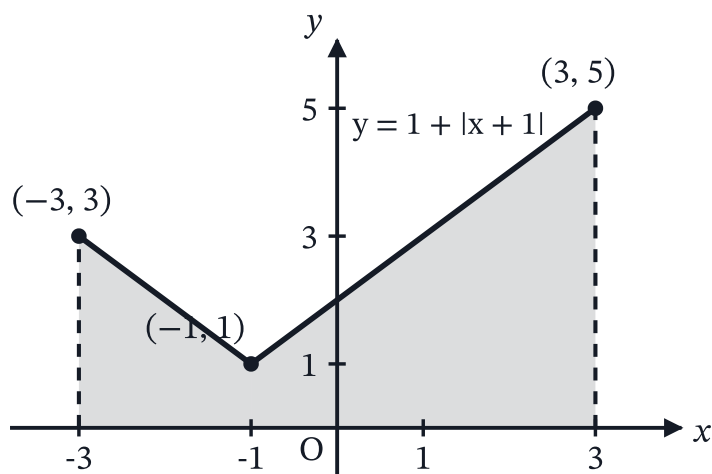
NOTE Check with the determinant formula: $\frac{1}{2} |-1(5 - 2) + 0(2 - 1) + 3(1 - 5)| = \frac{1}{2} |-3 - 12| = \frac{15}{2}$. ✓

ANSWER $\frac{15}{2}$ sq units

NCERT Exemplar Ch 8, Q23 — First Term Revision Paper

Draw a rough sketch of $y = 1 + |x + 1|$ and find the area of the region bounded by it, the lines $x = -3$, $x = 3$ and $y = 0$.

SOLUTION



The vertex sits at $(-1, 1)$, where the modulus changes definition — the integral splits there even though the graph never dips below the axis.

NOTE The graph is a V with its vertex where the modulus vanishes, at $x = -1$, $y = 1$. The two arms have slopes -1 (left) and $+1$ (right), and the whole curve lies above the x -axis.

1. Split at $x = -1$, where the definition changes.

2. For $-3 \leq x \leq -1$: $y = 1 - (x + 1) = -x$.

$$\int_{-3}^{-1} (-x) \, dx = \left[-\frac{x^2}{2} \right] \text{ from } -3 \text{ to } -1 = -\frac{1}{2} + \frac{9}{2} = 4$$

3. For $-1 \leq x \leq 3$: $y = 1 + (x + 1) = x + 2$.

$$\int_{-1}^3 (x + 2) \, dx = \left[\frac{x^2}{2} + 2x \right] \text{ from } -1 \text{ to } 3 = \left(\frac{9}{2} + 6 \right) - \left(\frac{1}{2} - 2 \right) = 12$$

4. Total area $= 4 + 12 = 16$.

NOTE Check geometrically: the region is two trapezia. Left, parallel sides 3 and 1 over width 2: $\frac{1}{2}(3 + 1)(2) = 4$. Right, parallel sides 1 and 5 over width 4: $\frac{1}{2}(1 + 5)(4) = 12$. ✓

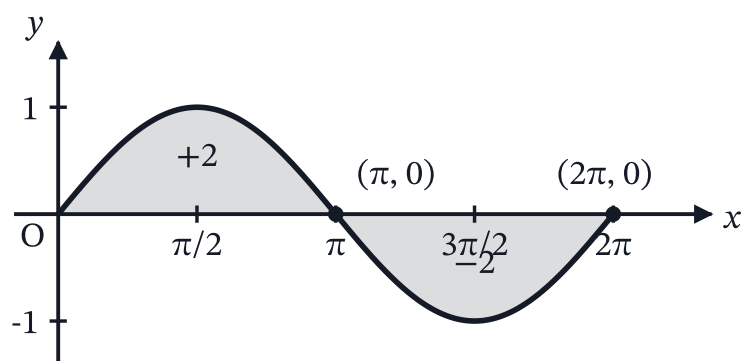
TRAP A modulus must be split at its zero even when the whole graph is above the axis — the **formula** changes there, not just the sign.

ANSWER 16 sq units

NCERT Exemplar Ch 8, Q17 — First Term Revision Paper

Find the area bounded by the curve $y = \sin x$ between $x = 0$ and $x = 2\pi$.

SOLUTION



The two lobes have equal area but opposite sign, which is why the plain integral is 0 and the **area** is 4.

NOTE 'Area bounded' means **area**, not the signed integral. $\sin x$ is negative on $(\pi, 2\pi)$, so the interval must be split.

1. $\int_0^{2\pi} \sin x \, dx = [-\cos x]$ from 0 to $2\pi = 0$ — the two halves cancel, so this is not the answer.
2. Split at π : $\int_0^{\pi} \sin x \, dx = [-\cos x]$ from 0 to $\pi = 1 + 1 = 2$.
3. $\int_{\pi}^{2\pi} \sin x \, dx = -1 - 1 = -2$, whose magnitude is 2.
4. Total area $= 2 + 2 = 4$.

TRAP Getting 0 is the sign that the modulus was forgotten. Any 'area' answer of zero for a curve that plainly encloses region is wrong.

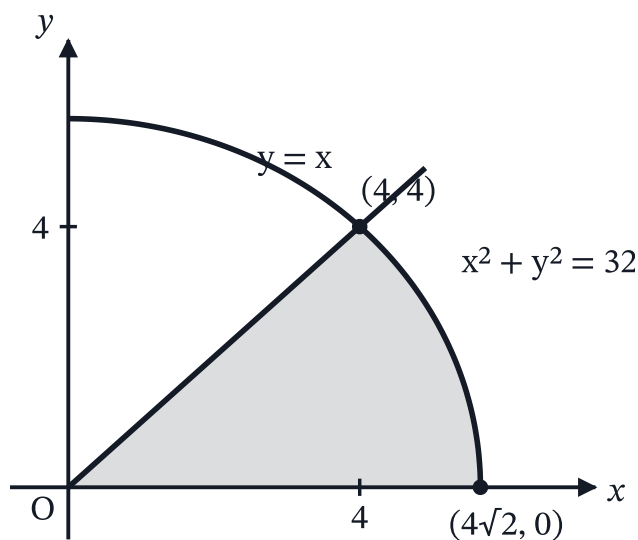
ANSWER 4 sq units

NCERT Exemplar Ch 8, Q27 — First Term Revision Paper

The area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$ is:

- (a) 4π sq units
- (b) 16π sq units
- (c) 32π sq units
- (d) 24 sq units

SOLUTION



The two boundaries through O are at 0 and $\pi/4$, so the region is a sector of angle $\pi/4$.

NOTE Before integrating, look at the shape: the two boundaries through the origin are the x-axis and the line $y = x$, at 0 and $\frac{\pi}{4}$. The region is a **circular sector**.

1. The circle has radius $\sqrt{32} = 4\sqrt{2}$, so $r^2 = 32$.
2. Sector area $= \frac{1}{2} r^2 \theta = \frac{1}{2} \times 32 \times \frac{\pi}{4} = 4\pi$.

NOTE By integration the same answer comes from $\int_0^4 x \, dx + \int_4^{4\sqrt{2}} \sqrt{32 - x^2} \, dx = 8 + (4\pi - 8) = 4\pi$, which is a useful check but much longer.

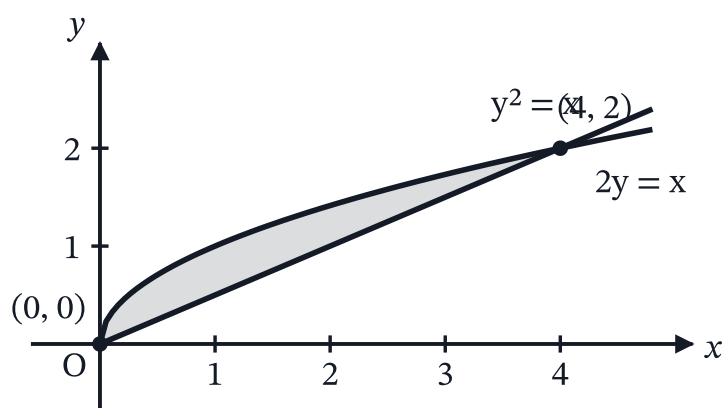
ANSWER 4π sq units

NCERT Exemplar Ch 8, Q29 — First Term Revision Paper

The area of the region bounded by the parabola $y^2 = x$ and the straight line $2y = x$ is:

- (a) $\frac{4}{3}$ sq units
- (b) 1 sq unit
- (c) $\frac{2}{3}$ sq units
- (d) $\frac{1}{3}$ sq units

SOLUTION



Integrating along y : the line $x = 2y$ is to the right of the parabola $x = y^2$ for $0 < y < 2$.

1. Integrating along y is easier here, since both curves give x in terms of y : $x = y^2$ and $x = 2y$.
2. They meet where $y^2 = 2y$, i.e. $y = 0$ and $y = 2$.
3. On $0 < y < 2$ the line is to the right ($2y > y^2$), so

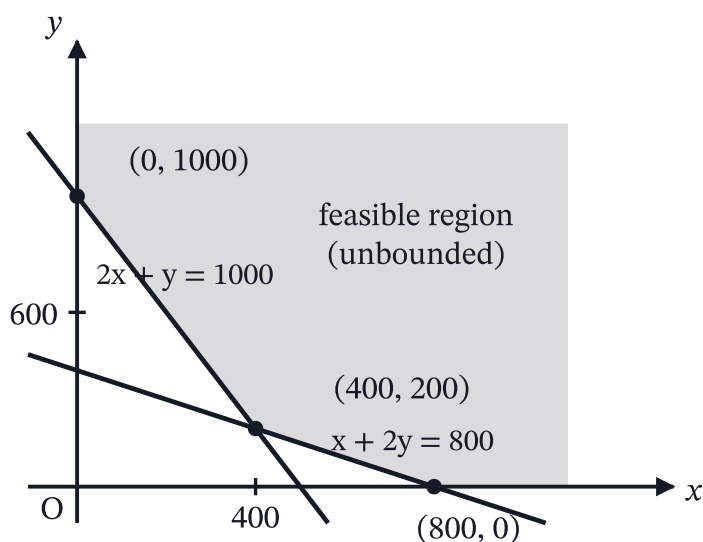
$$\text{Area} = \int_0^2 (2y - y^2) dy = \left[y^2 - \frac{y^3}{3} \right] \text{ from } 0 \text{ to } 2 = 4 - \frac{8}{3} = \frac{4}{3}$$

NOTE Choosing the variable of integration well is half the work: doing this one in x needs the region split at $x = 0$ and a square root.

ANSWER $\frac{4}{3}$ sq units

Every gram of rice provides 0.1 g of protein and 0.25 g of carbohydrates; for wheat the values are 0.05 g and 0.5 g. Rice costs ₹40 per kg and wheat ₹60 per kg. The minimum daily requirements are 50 g of protein and 200 g of carbohydrates. In what quantities should they be mixed to meet the requirements at least cost? Formulate and solve graphically.

SOLUTION



The region is unbounded upwards, so the corner value 28 has to be confirmed:
 $0.04x + 0.06y < 28$ misses the region entirely.

1. **Formulation.** Let the diet contain x g of rice and y g of wheat.

$$\text{Protein: } 0.1x + 0.05y \geq 50 \Rightarrow 2x + y \geq 1000$$

$$\text{Carbohydrate: } 0.25x + 0.5y \geq 200 \Rightarrow x + 2y \geq 800$$

2. Cost in rupees (₹40 per 1000 g is ₹0.04 per gram): minimise $C = 0.04x + 0.06y$, with $x, y \geq 0$.
3. **Corner points.** The region is unbounded, above both lines. $2x + y = 1000$ meets the y -axis at $(0, 1000)$; $x + 2y = 800$ meets the x -axis at $(800, 0)$.
4. The two lines cross where $y = 1000 - 2x$ gives $x + 2000 - 4x = 800$, so $x = 400$ and $y = 200$.
5. $C(0, 1000) = ₹60$; $C(800, 0) = ₹32$; $C(400, 200) = 16 + 12 = ₹28$.
6. **Unbounded check.** The smallest corner value is 28; the open half plane $0.04x + 0.06y < 28$ has no point in common with the feasible region, so 28 really is the minimum.

NOTE 400 g of rice and 200 g of wheat — check the requirements: protein $40 + 10 = 50$ g ✓, carbohydrate $100 + 100 = 200$ g ✓. Both bind exactly, which is why this corner is optimal.

TRAP An unbounded feasible region needs that last check. Quoting the smallest corner value without it is incomplete, because a minimum need not exist.

ANSWER 400 g of rice and 200 g of wheat, at a minimum cost of ₹28

PB Q13(b) — Heritage 2016–17 Pre-Board

Every gram of wheat provides 0.1 g of protein and 0.25 g of carbohydrates; every gram of rice provides 0.05 g of protein and 0.5 g of carbohydrates. Wheat costs ₹5 per kg and rice ₹20 per kg. The minimum daily requirements are 50 g of protein and 200 g of carbohydrates. In what quantities should they be mixed to meet the requirements at least cost?

SOLUTION

NOTE The same problem as Q22 of the 2024-25 First Term paper, with the two grains exchanged and the prices changed. Working both is the best test of whether you can formulate rather than recall.

1. Let the diet contain x g of wheat and y g of rice.

$$\text{Protein: } 0.1x + 0.05y \geq 50 \Rightarrow 2x + y \geq 1000$$

$$\text{Carbohydrate: } 0.25x + 0.5y \geq 200 \Rightarrow x + 2y \geq 800$$

2. Cost per gram is ₹0.005 for wheat and ₹0.02 for rice, so minimise $C = 0.005x + 0.02y$.
3. **Corner points** of the unbounded region: (0, 1000), (400, 200) and (800, 0).
4. $C(0, 1000) = ₹20$; $C(400, 200) = 2 + 4 = ₹6$; $C(800, 0) = ₹4$.
5. **Unbounded check.** The smallest is 4; and $0.005x + 0.02y < 4$ has no point in common with the region, so 4 is genuinely the minimum.

NOTE The answer is all wheat — 800 g of it, no rice — because wheat is four times cheaper per gram and carries twice the protein. Check the requirements: protein 80 g ≥ 50 ✓, carbohydrate 200 g ≥ 200 ✓, the second binding exactly.

TRAP With the First Term paper's numbers the optimum is an interior corner; here it slides onto an axis. Never assume the answer is the intersection of the two constraint lines — test every corner.

ANSWER 800 g of wheat and no rice, at a minimum cost of ₹4

NCERT Exemplar Ch 12, Q23 — NCERT Exemplar

A person wishes to take tablets X and Y to meet iron, calcium and vitamin requirements at least cost.

SOLUTION

1. With x tablets of X and y of Y: $6x + 2y \geq 18$, $3x + 3y \geq 21$, $2x + 4y \geq 16$; minimise $2x + y$.
2. Corner points $(0, 9)$, $(1, 6)$, $(6, 1)$, $(8, 0)$ give 9, 8, 13, 16.
→ 1 tablet of X and 6 of Y, at a cost of ₹8

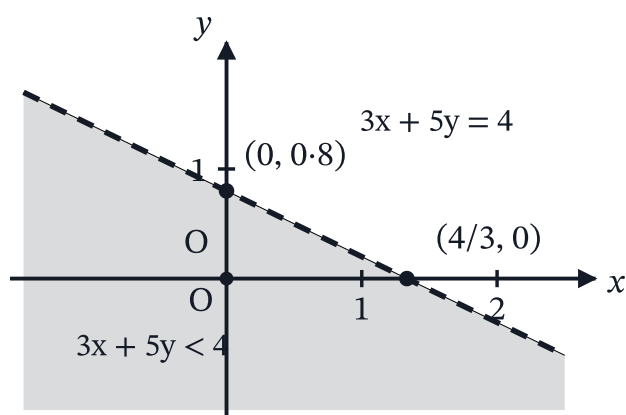
WHY IT IS SET HERE *The same formulation with three constraints.*

FT Q19(ii) — Heritage 2024–25 First Term

The solution set of the inequation $3x + 5y < 4$ is:

- (a) an open half plane containing the origin
- (b) an open half plane not containing the origin
- (c) the whole xy -plane except the points on the line $3x + 5y = 4$
- (d) none of these

SOLUTION



The boundary is dashed because the inequality is strict; the origin satisfies it, so the shaded side is the one containing O.

1. A strict linear inequality in two variables describes an **open** half plane — the boundary line is excluded.
2. Test the origin: $3(0) + 5(0) = 0 < 4$ is true.
3. So the origin lies in the solution set, and the half plane is the one containing it.

NOTE The origin is the quickest test point whenever the line does not pass through it. If it did, use any other convenient point.

ANSWER an open half plane containing the origin

NCERT Exemplar Ch 12 — NCERT Exemplar

Which half plane does a given strict linear inequality describe?

SOLUTION

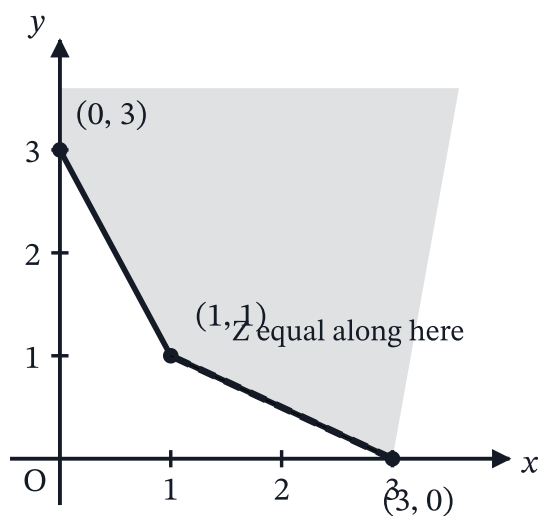
1. A linear inequality always describes a half plane; a **strict** one leaves the boundary line out.
2. Substitute the origin: whichever way the inequality falls tells you which side is meant.
→ The open half plane on the origin's side, when the origin satisfies it

WHY IT IS SET HERE *Test the origin; a strict inequality leaves the boundary out.*

The corner points of the feasible region determined by a system of linear constraints are $(0, 3)$, $(1, 1)$ and $(3, 0)$. Let $Z = px + qy$ with $p, q > 0$. Find the condition on p and q for the minimum of Z to occur at both $(3, 0)$ and $(1, 1)$.

- (a) $q = 2p$
- (b) $p = 2q$
- (c) $p = q$
- (d) $p = 3q$

SOLUTION



Z takes the same value at $(3, 0)$ and $(1, 1)$, so the objective line is parallel to that edge and every point of it is optimal.

1. If the minimum occurs at two corner points, Z takes the same value at both.
2. $Z(3, 0) = 3p$ and $Z(1, 1) = p + q$.
3. Setting them equal: $3p = p + q$, so $q = 2p$.

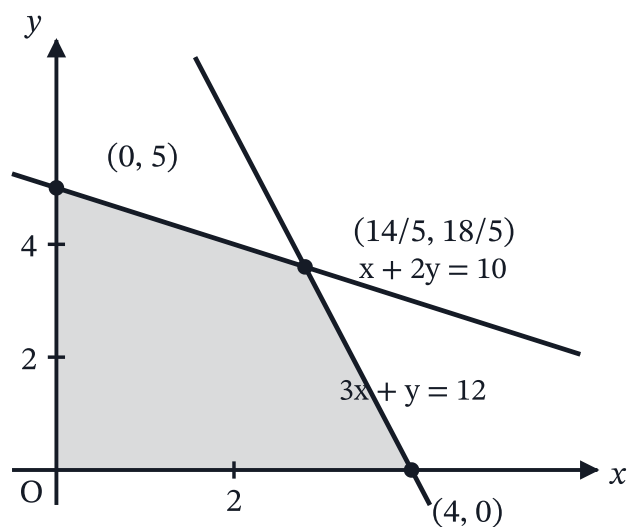
NOTE When two corner points tie, every point on the segment joining them is also optimal — the objective line is parallel to that edge, and the problem has infinitely many optimal solutions.

ANSWER $q = 2p$

The number of feasible solutions of the linear programming problem: maximise $Z = 15x + 30y$ subject to $3x + y \leq 12$, $x + 2y \leq 10$, $x \geq 0$, $y \geq 0$ is:

- (a) infinite
- (b) 1
- (c) 2
- (d) 4

SOLUTION



Every point of the shaded quadrilateral is a feasible solution, so there are infinitely many — only the *optimum* is confined to a corner.

1. A **feasible solution** is any point of the feasible region, not just a corner point.
2. The constraints define a bounded quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(\frac{14}{5}, \frac{18}{5})$ and $(0, 5)$ — a region with area, not a finite set of points.
3. Every point of it satisfies all four constraints, so there are infinitely many feasible solutions.

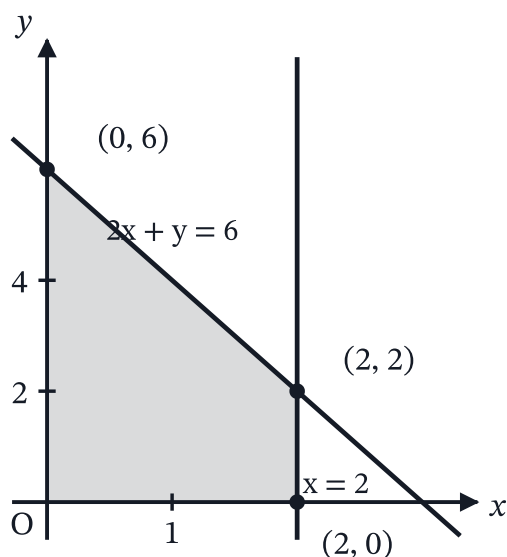
TRAP The question asks for feasible **solutions**, not corner points or optimal solutions. Only the optimal value is unique here; the feasible set never is, unless it collapses to a point.

ANSWER infinite

NCERT Exemplar Ch 12, Q1 — First Term Revision Paper

Determine the maximum value of $Z = 11x + 7y$ subject to $2x + y \leq 6$, $x \leq 2$, $x \geq 0$, $y \geq 0$.

SOLUTION



Maximum 42 at $(0, 6)$: the constraint charges twice as much for a unit of x , and $11 < 2 \times 7$.

1. Corner points of the feasible region: $(0, 0)$; $(2, 0)$ where $x = 2$ meets the x -axis; $(2, 2)$ where $x = 2$ meets $2x + y = 6$; and $(0, 6)$ where $2x + y = 6$ meets the y -axis.
2. $Z(0, 0) = 0$.
3. $Z(2, 0) = 22$.
4. $Z(2, 2) = 22 + 14 = 36$.
5. $Z(0, 6) = 42$.

NOTE The maximum is 42 at $(0, 6)$, even though x has the larger coefficient: the constraint $2x + y \leq 6$ charges twice as much for a unit of x , and $11 < 2 \times 7$.

ANSWER Maximum $Z = 42$ at $(0, 6)$

NCERT Exemplar Ch 12, Q10 — NCERT Exemplar

From a drawn feasible region, determine the maximum and minimum of $Z = x + 2y$.

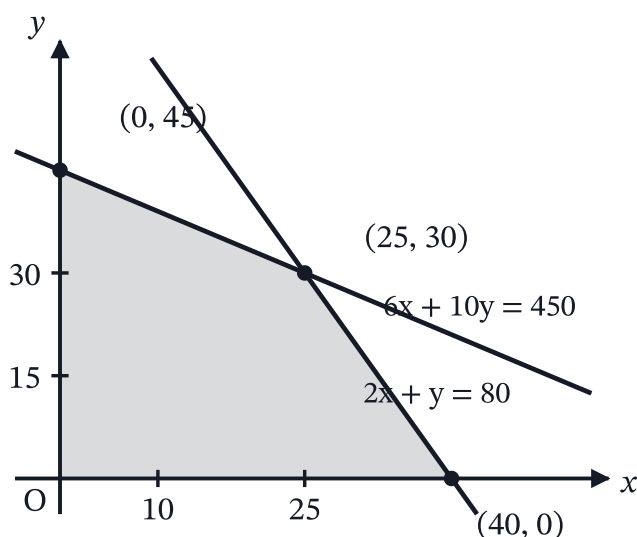
SOLUTION

1. Read the vertices off the figure, evaluate Z at each, and take the largest and smallest.
 2. If the region is unbounded, a corner value is only optimal once the open half plane beyond it is checked for points of the region.
- The optimum is always at a corner point of a bounded region

WHY IT IS SET HERE *Same corner-point method; the Exemplar gives the region as a figure rather than as a list of corners.*

A manufacturer produces Model X and Model Y bikes. Model X takes 6 man-hours per unit and Model Y 10; 450 man-hours are available per week. Handling and marketing cost ₹2000 per unit of X and ₹1000 per unit of Y, and ₹80,000 is available per week. Profits are ₹1000 and ₹500 per unit. How many of each should be produced for maximum profit, and what is that profit?

SOLUTION



$P = 40,000$ at both $(40, 0)$ and $(25, 30)$, so the objective line is parallel to $2x + y = 80$ and the whole edge between them is optimal.

1. **Formulation.** Let x units of X and y units of Y be made per week.

$$\text{Labour: } 6x + 10y \leq 450 \quad \text{Funds: } 2000x + 1000y \leq 80000 \Rightarrow 2x + y \leq 80$$

2. Maximise $P = 1000x + 500y$, $x, y \geq 0$.

3. **Corner points.** $(0, 0)$; on the x -axis the binding limit is $2x \leq 80$, giving $(40, 0)$; on the y -axis it is $10y \leq 450$, giving $(0, 45)$.

4. The two lines meet where $y = 80 - 2x$ gives $6x + 800 - 20x = 450$, so $14x = 350$, $x = 25$ and $y = 30$.

5. **Profits.** $P(0,0) = 0$; $P(40, 0) = 40000$; $P(25, 30) = 25000 + 15000 = 40000$; $P(0, 45) = 22500$.

NOTE Two corner points tie at ₹40,000, so **every** point of the edge joining $(40, 0)$ and $(25, 30)$ is optimal — the objective line $1000x + 500y$ is parallel to $2x + y = 80$. Any whole-number point on that edge, such as $(30, 20)$ or $(35, 10)$, earns the same ₹40,000.

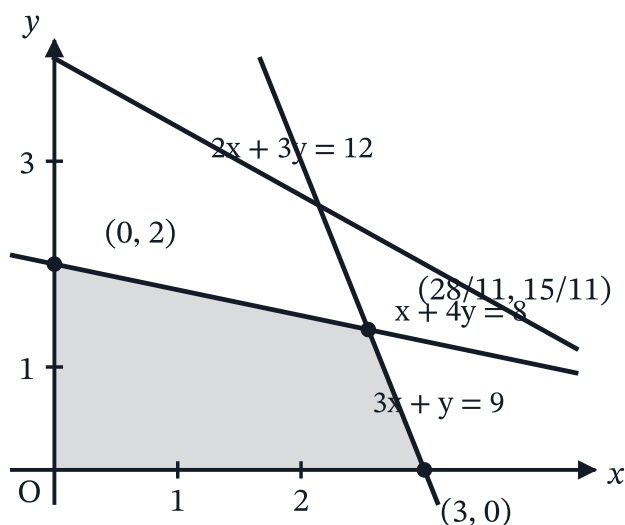
TRAP Reporting only one of the two corner points is not wrong, but missing the tie misses what the question is really showing.

ANSWER Maximum profit ₹40,000 — at $(40, 0)$, at $(25, 30)$, and at every point of the edge between them

NCERT Exemplar Ch 12, Q21 — First Term Revision Paper

Maximise $Z = x + y$ subject to $x + 4y \leq 8$, $2x + 3y \leq 12$, $3x + y \leq 9$, $x \geq 0$, $y \geq 0$.

SOLUTION



$2x + 3y = 12$ never touches the region — it is redundant. The other two pairs of lines meet outside it.

1. Test each pair of lines and keep only the intersections that satisfy every constraint.
2. $x + 4y = 8$ with $3x + y = 9$: $x = 8 - 4y$ gives $24 - 12y + y = 9$, so $y = \frac{15}{11}$ and $x = \frac{28}{11}$.
Check $2x + 3y = \frac{101}{11} \leq 12$ ✓.
3. $x + 4y = 8$ with $2x + 3y = 12$ gives $(4.8, 0.8)$, but there $3x + y = 15.2 > 9$ ✗ — infeasible.
4. $2x + 3y = 12$ with $3x + y = 9$ gives $(\frac{15}{7}, \frac{18}{7})$, but there $x + 4y = \frac{87}{7} > 8$ ✗ — infeasible.
5. So the corner points are $(0, 0)$, $(3, 0)$, $(\frac{28}{11}, \frac{15}{11})$ and $(0, 2)$.
6. Z : 0 , 3 , $\frac{43}{11} \approx 3.91$, and 2 .

NOTE The constraint $2x + 3y \leq 12$ turns out to be redundant — it is satisfied automatically at every corner. Spotting that early saves work, but it can only be confirmed after the corners are found.

TRAP Do not assume every pair of lines gives a corner. Two of the three pairs here meet outside the feasible region.

ANSWER Maximum $Z = \frac{43}{11}$ at $(\frac{28}{11}, \frac{15}{11})$

Companion to *School Paper Based*. Every concept here is one that a question in that deck actually uses.